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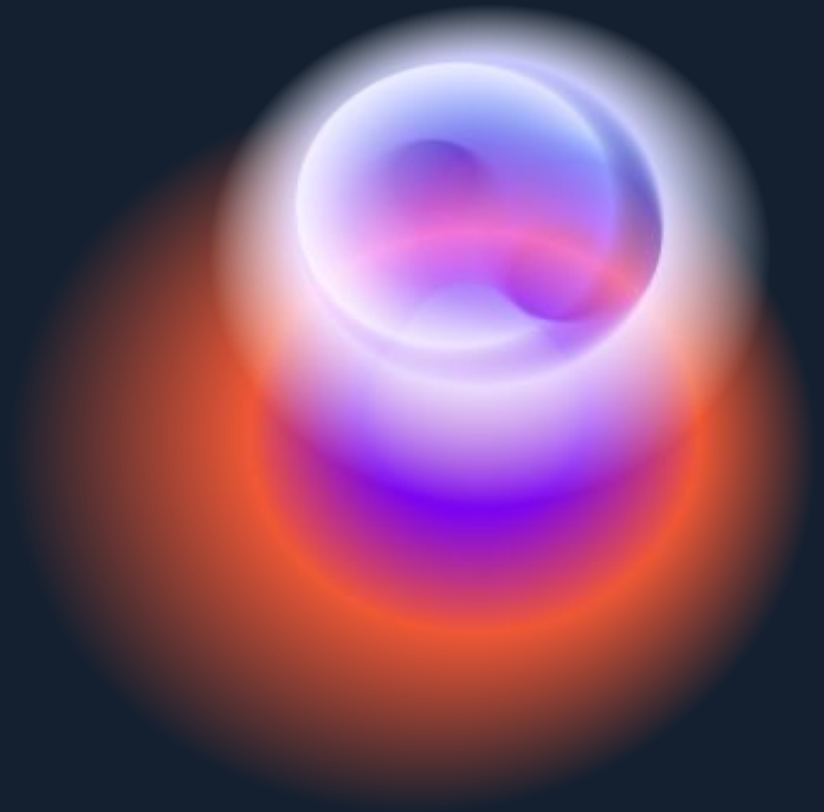
Harnesses: Beyond the Model

Same model. Different agent.

Dr. João Carvalho
Uthereal · CAS AI in Industry, ETH Zürich
Autumn 2026

Follow the presentation here





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The Team

A proven team with deep expertise in AI research, product delivery, and go-to-market, supported by world-class advisors and investors.

Founders



Wael, CEO

- Cloud AI Architect - Healthcare and Life Sciences
- Published, multiple award winner
- Ex-AWS - Led sales and delivery of $80m+ in AI
- 14 years in leading Data Analytics & AI teams
- MSc in Information Security & AI



João, CTO

- PhD @ ETH Zürich in Machine Learning
- Associate researcher @ ETH AI Center
- Expert in Robust ML
- MSc in AI Biomedical Engineering

Team



Carly, Chief Growth Officer

10+ years in go-to-market and sales



Victor, Founding AI Engineer

Top 1% ETH data scientist



Riccardo, AI Engineer

ETH data scientist



Ozgur, AI Engineer

ETH GenAI Specialist



Daniela, Project Management

PM specialist

Rodrigo, UX specialist

UI, UX FullStack

Advisors & Mentors



Paul H. Müller

CTO and Co-founder of Adjust (1.2B exit)
EWOR partner



Christian Haase

CEO Quintessenz Verlags GmbH
20+years in medical publishing



Andreas Slotosch

CGO and Co-founder of Beekeeper



Julia Vogt, Prof. Dr.

Leading Medical Data Science Group, ETH Zurich



Bernd Stadlinger Prof. Dr.

Expert in AI applied in Medicine

Spin-off | ETH zürich

Microsoft Azure

aws

Universität Zürich[VZH]

EWOR

Spot the harness

Four products on the same families of models. What is different?

Chat assistant

Answers in one go

Sees your message
Touches nothing
One turn, then stops

Deep research

Searches, reads, cites

Sees the web
Touches nothing
Many turns, then stops

Coding agent

Edits, runs tests, retries

Sees your files
Changes them
Retries until tests pass

Support bot

Looks up, then acts

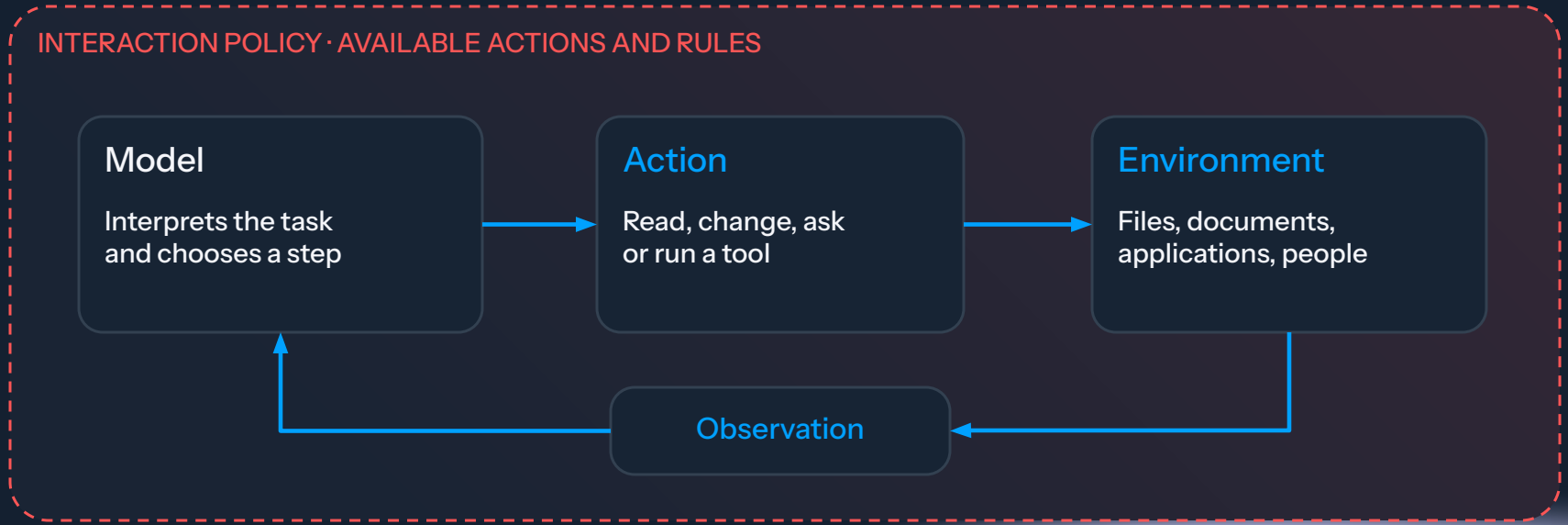
Sees your account
Acts in your name
Stops for approval

Show of hands: which of these could notice it was wrong, and try again?

Audience exercise. Product categories, not benchmark results; vendors change the models behind them.

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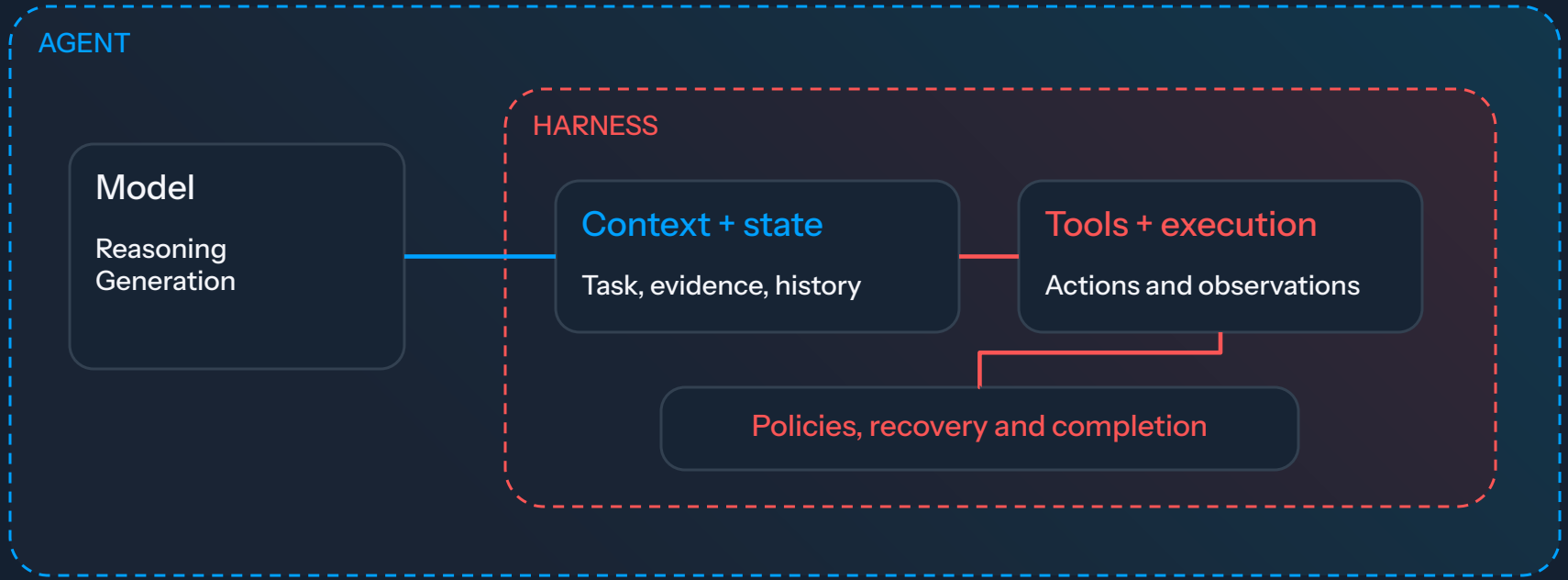
Agency means being able to act



[4] Building effective agents

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Agent = model + harness



The harness supplies the working environment and controls how work proceeds.

[1] Anatomy of coding-agent harnesses

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What a harness is not

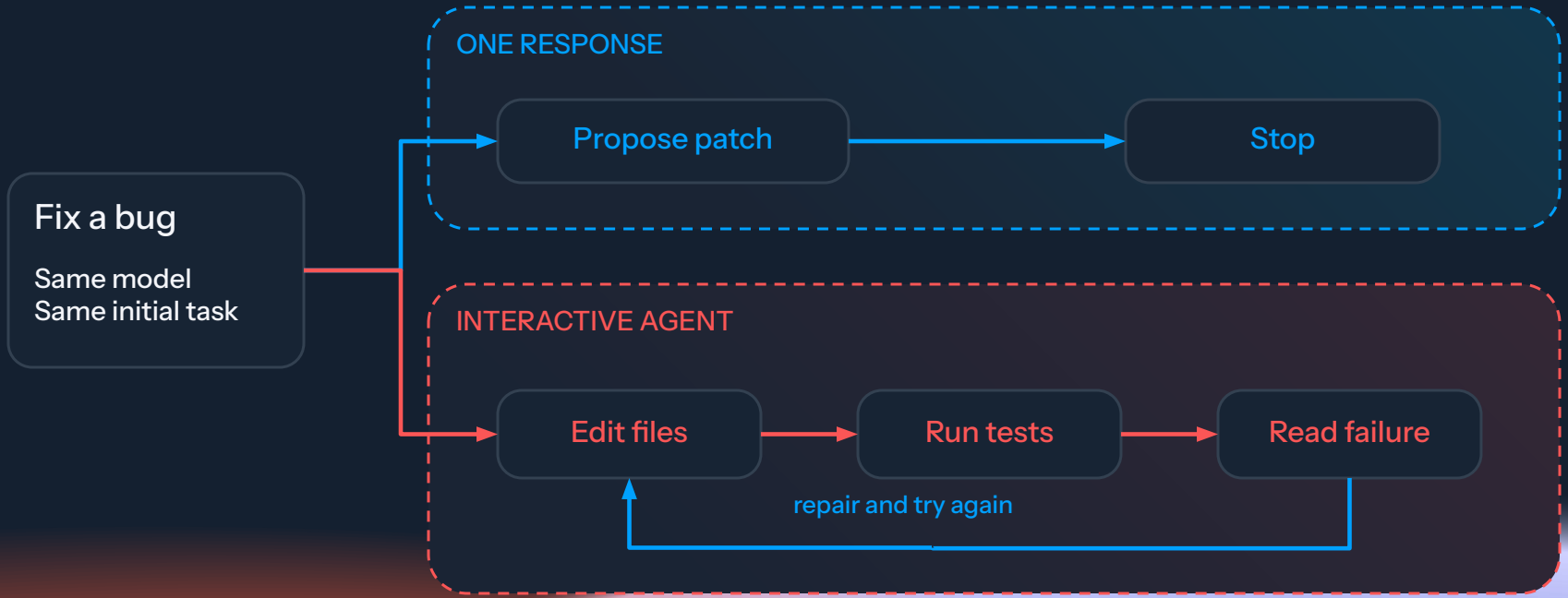


These roles can coexist. None alone describes the whole operating system.

[1] Anatomy of coding-agent harnesses / [4] Building effective agents

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Same task, different ability to respond

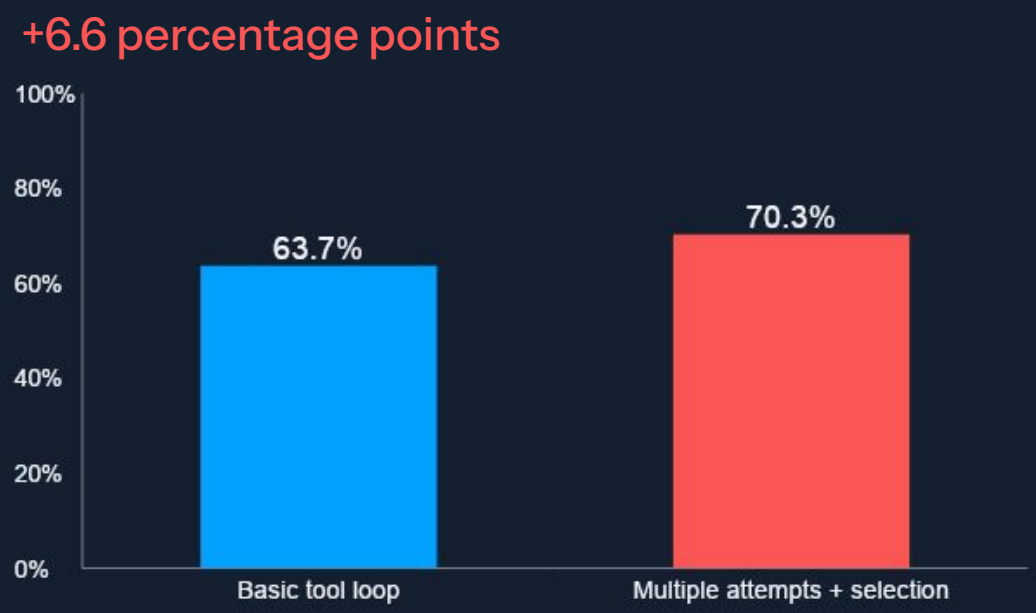


Audience question: which setup can discover that its first patch fails a test?

[4] Building effective agents

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The same model can solve more tasks



WHAT CHANGED

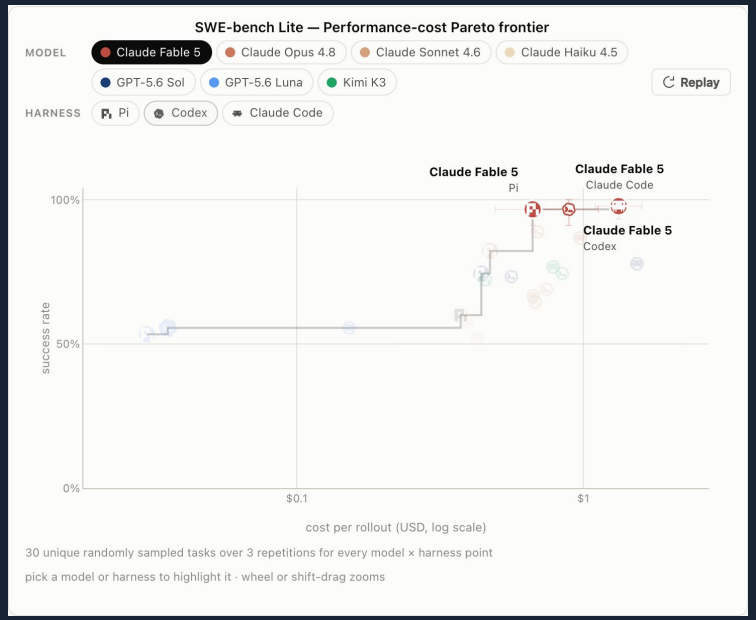
- Multiple attempts
- Regression-test filter
- Candidate selection

Same Claude 3.7 Sonnet, same 489 tasks. More inference compute. Vendor-reported SWE-bench result.

[2] Claude 3.7 Sonnet evaluation

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Similar success can cost twice as much



SAME MODEL: CLAUDE FABLE 5

CLAUDE CODE

$1.33

97.8% success

PI

$0.67

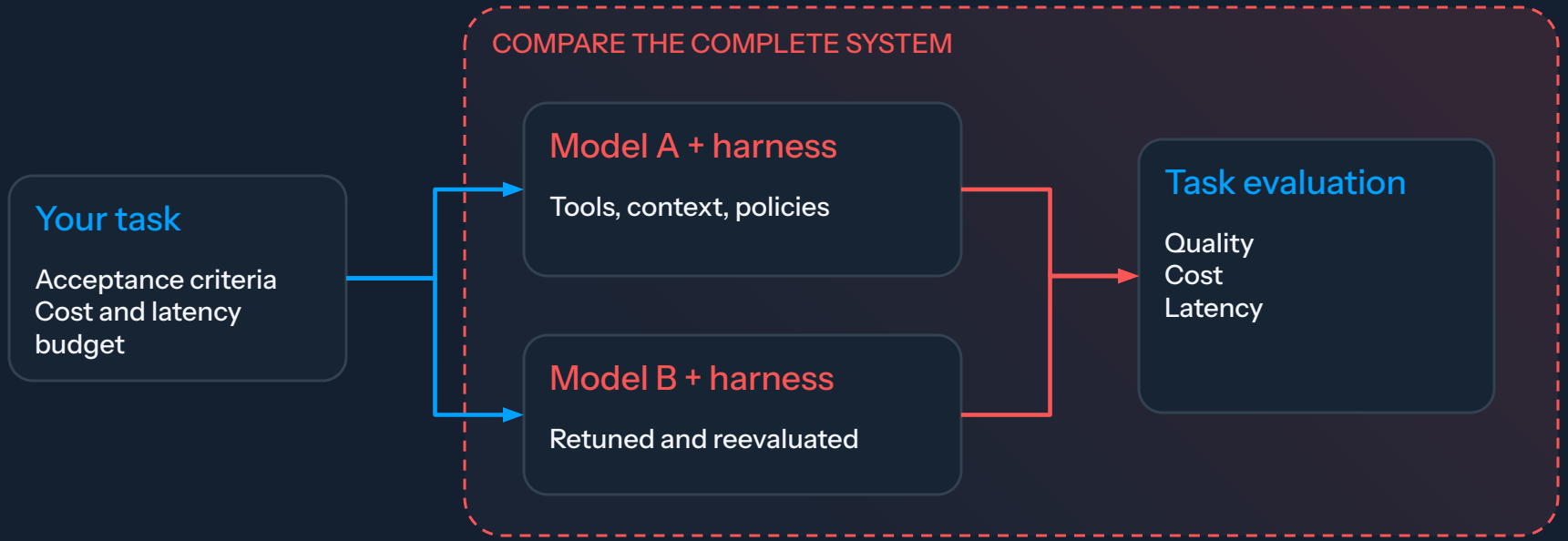
96.7% success

SWE-bench Lite. 30 sampled tasks × 3 runs. Mean API cost per rollout. Small sample, not proof of equivalence.

[3] HarnessTax results and methodology

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Capability and cost are design choices

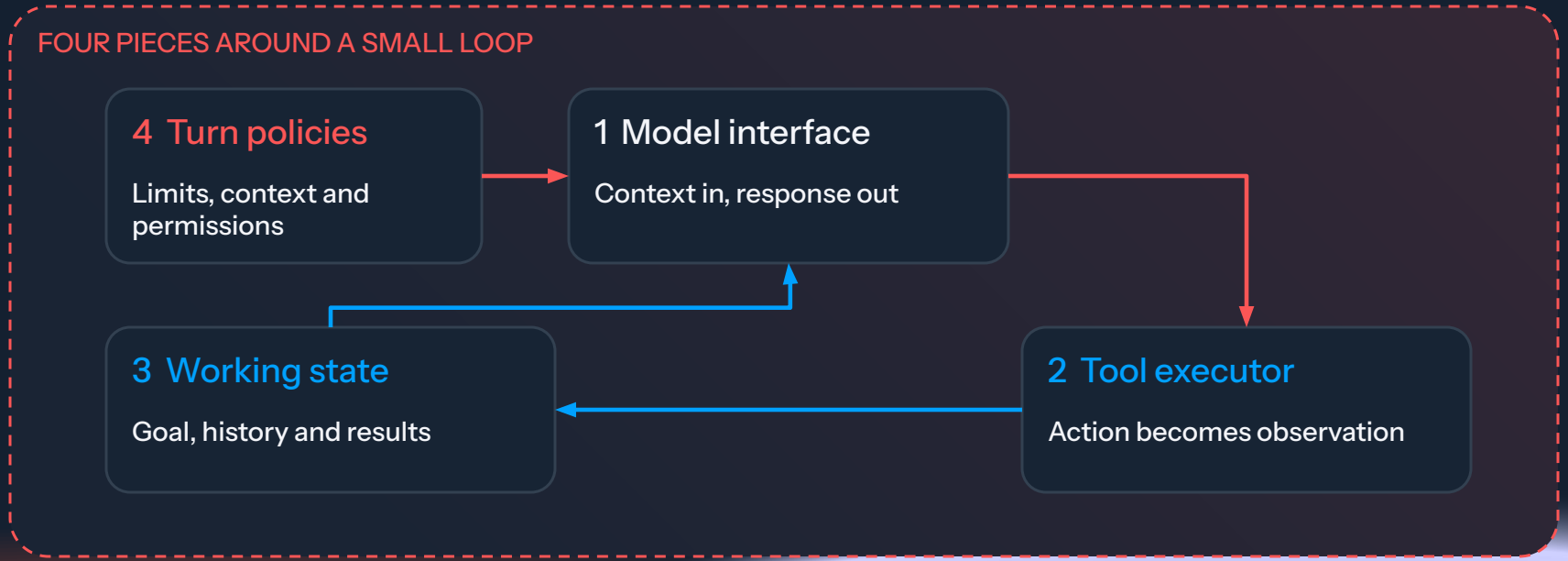


A shared model interface makes switching easier. Evaluation makes it meaningful.

[1] Anatomy of coding-agent harnesses / [3] HarnessTax results and methodology

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The minimum harness



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Three rules of a harness

1 The model proposes, the harness decides

Every action passes a checked executor: arguments, permissions, timeout. The model never touches the world directly.

2 “Done” is a check, not a claim

A completion check decides whether the task is finished. If it fails, the feedback goes back in for another turn.

3 A limit is a stop, not a success

Turn, cost or time limits return “stopped”, with the reason. A stop is never reported as a result.

ENFORCED BY SOFTWARE, NOT BY PROMPTS

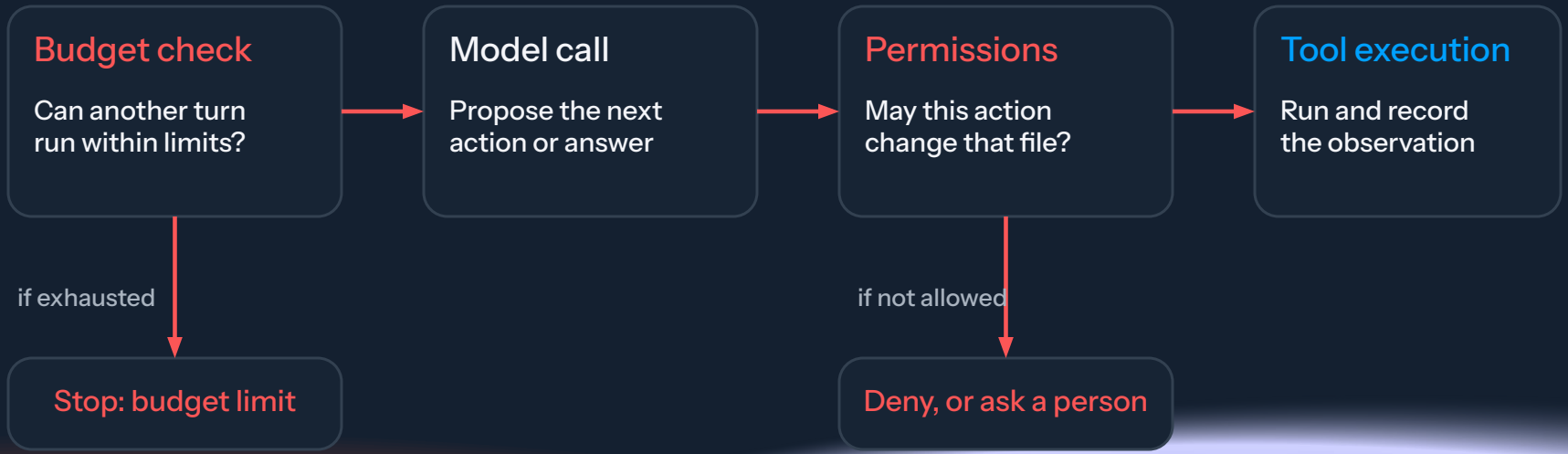
An instruction asks the model to behave. A harness rule holds whatever the model does.

execute_checked(action) · completion_check(response) · stopped_with_reason(state)

[1] Harness Engineering: anatomy of coding-agent harnesses

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Where humans keep control



Approvals, read-only modes, budgets and logs live in the harness, as software around every turn.

[1] Anatomy of coding-agent harnesses

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The harness is becoming its own industry

YC Fall 2026: four startups building the systems around the models



Agent Relay

Coordination

Shared context and resumable agent workflows



herdr

Agent runtime

Persistent sessions across machines and clients



Agent 37

Execution environment

Persistent sandboxes with tools and integrations



Papaya

Harness optimization

Production traces become proposed fixes and PRs

[1]: Agent Relay / herdr / Agent 37 / Papaya

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Your questions

Pick one. Each card opens a short deep dive and brings us back here.

Q1
Isn't it just the prompt?
4 slides

Q2
What is actually inside the loop?
3 slides

Q3
Why does context matter so much?
4 slides

Q4
Can it remember, and work for days?
4 slides

Q5
Are more agents better?
5 slides

Q6
What does it cost, and how do we judge it?
4 slides

Q7
Can I trust it with my data and tools?
4 slides

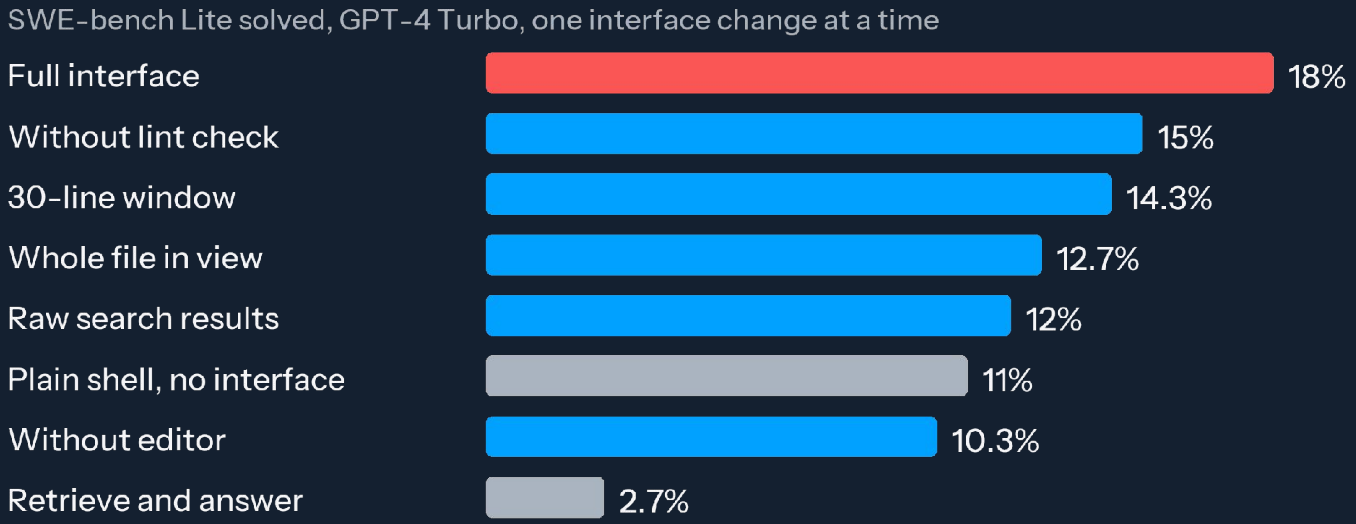
Q8
Where is this going?
5 slides

Deep dives: Q1 → 16–19 · Q2 → 20–22 · Q3 → 23–26 · Q4 → 27–30 · Q5 → 31–35 · Q6 → 36–39 · Q7 → 40–43 · Q8 → 44–48

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The interface is the agent's world

SWE-agent: Yang et al. NeurIPS 2024 · ablations, GPT-4 Turbo on SWE-bench Lite



Same model, same tasks: each interface change moves the score.

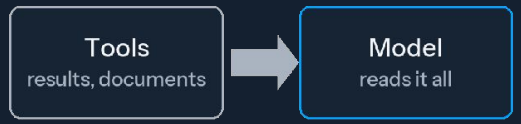
[5] SWE-agent (NeurIPS 2024), Tables 1 and 3

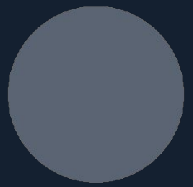
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Tools are written for the model, not for people

Code execution with MCP · Writing effective tools for agents · Anthropic Engineering · 2025

Every tool result goes through the model





150,000 tokens
the model has to read

Tools as code, results filtered in a sandbox



2,000 tokens
the model has to read
75× fewer

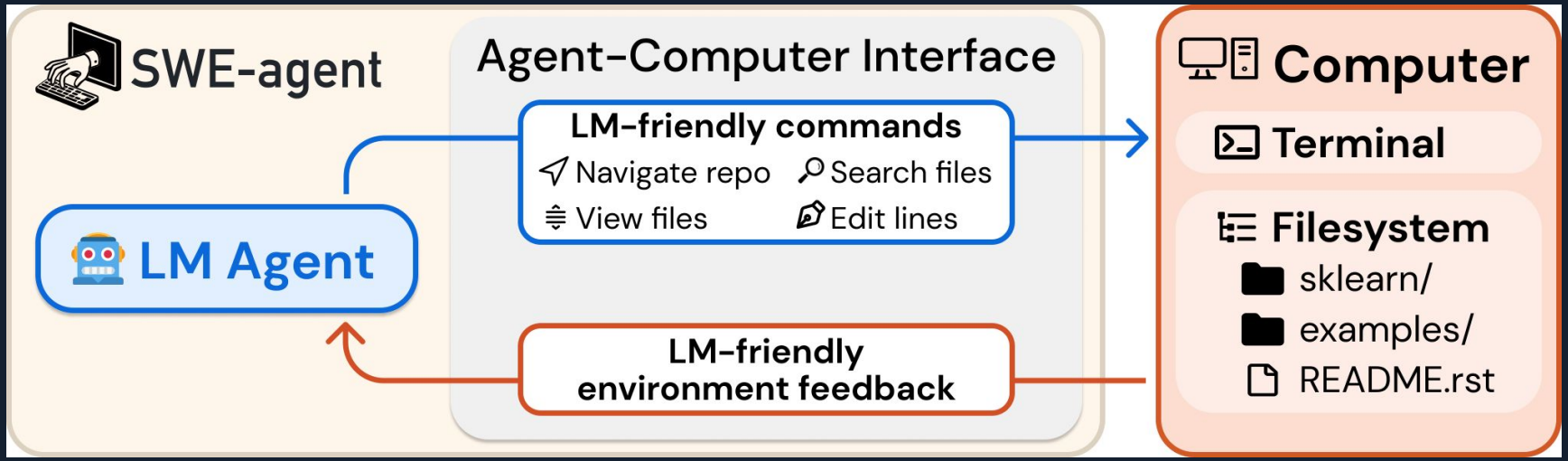
A tool description is a prompt; a tool result is context. Both are paid for in tokens.

[27] Anthropic Engineering, 2025: one worked example, not an average

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What the agent sees: SWE-agent's interface

SWE-agent · Yang et al. · NeurIPS 2024 · Figure 1



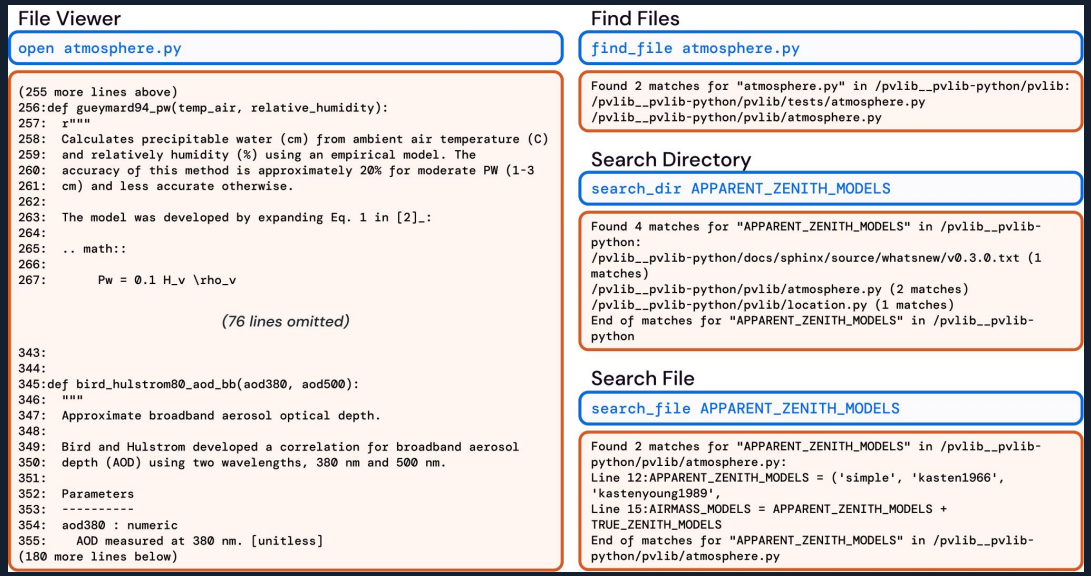
The language model never touches the computer directly: every command and every observation passes through the agent-computer interface, which the authors designed for a model rather than for a person.

[5] Yang et al. · SWE-agent · Fig. 1 · reproduced from the arXiv version for discussion

← your questions

The file viewer and the search commands

SWE-agent · Yang et al. · NeurIPS 2024 · Figure 10



File viewer

A window of at most 100 lines.

Search

Capped, summarised results.

The two components behind the ablation numbers.

An open harness, part by part

OpenHands (ICLR 2025): the platform behind a family of coding agents

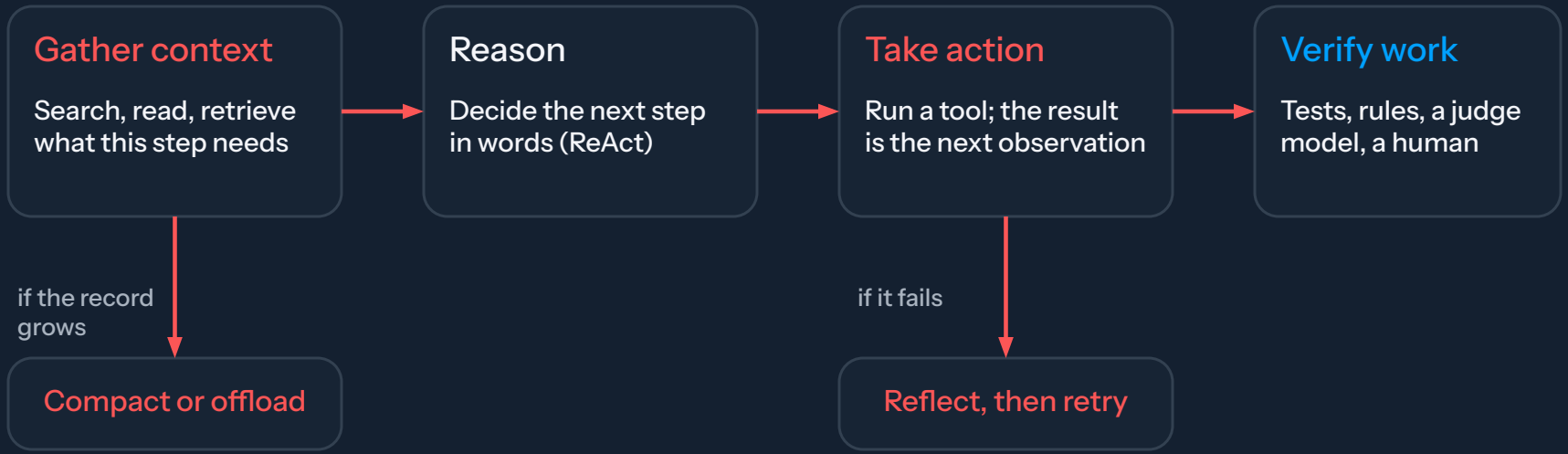
COMPONENT	WHAT IT DOES	IN THIS DECK
Agent	A step function: current state in, next action out	Slide 11
Event stream	The chronological record of past actions and observations	Q3
Runtime	An isolated Docker sandbox in which every action is executed	Rule 1, slide 13
AgentSkills	Tools as a Python package of utility functions	Q1
Delegation	An agent hands a subtask to another agent	Q5
Evaluation	15 established benchmarks integrated into the platform	Q6

The anatomy of slide 11, written down as a peer-reviewed system.

[6] OpenHands: an open platform for AI software developers as generalist agents (ICLR 2025)

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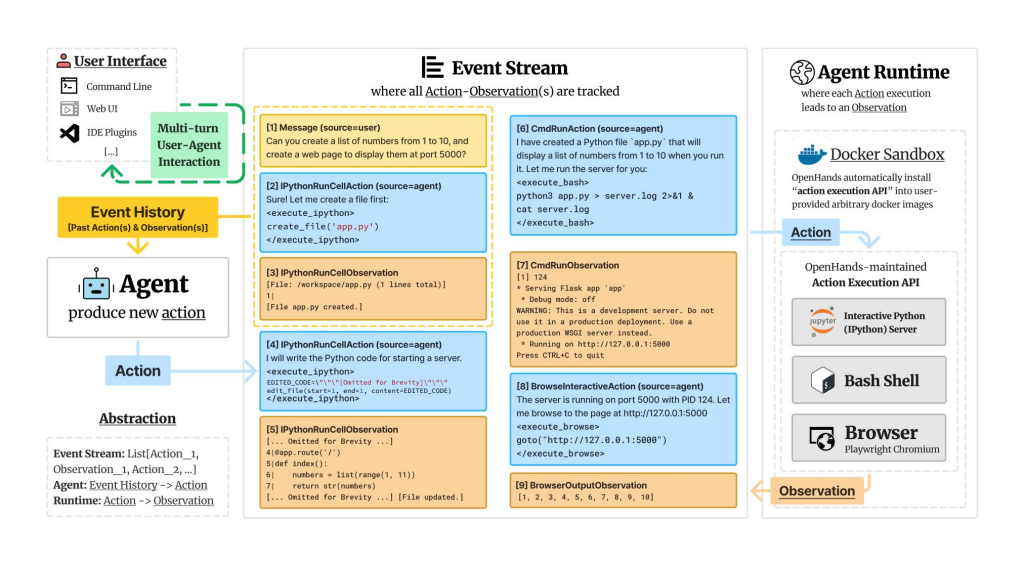
Gather context, act, verify, repeat



Anthropic's loop for Claude Code: "gather context → take action → verify work → repeat." ReAct (ICLR 2023) and Reflexion (NeurIPS 2023) are its peer-reviewed ancestors: HumanEval 80% → 91% with reflection.

OpenHands: the architecture in one figure

OpenHands · Wang et al. · ICLR 2025 · Figure 2



Agent and event stream

The agent produces the next action from the event stream, the record of every action and observation.

Runtime

Actions run in a Docker sandbox with a Python server, a shell and a browser; results come back as observations.

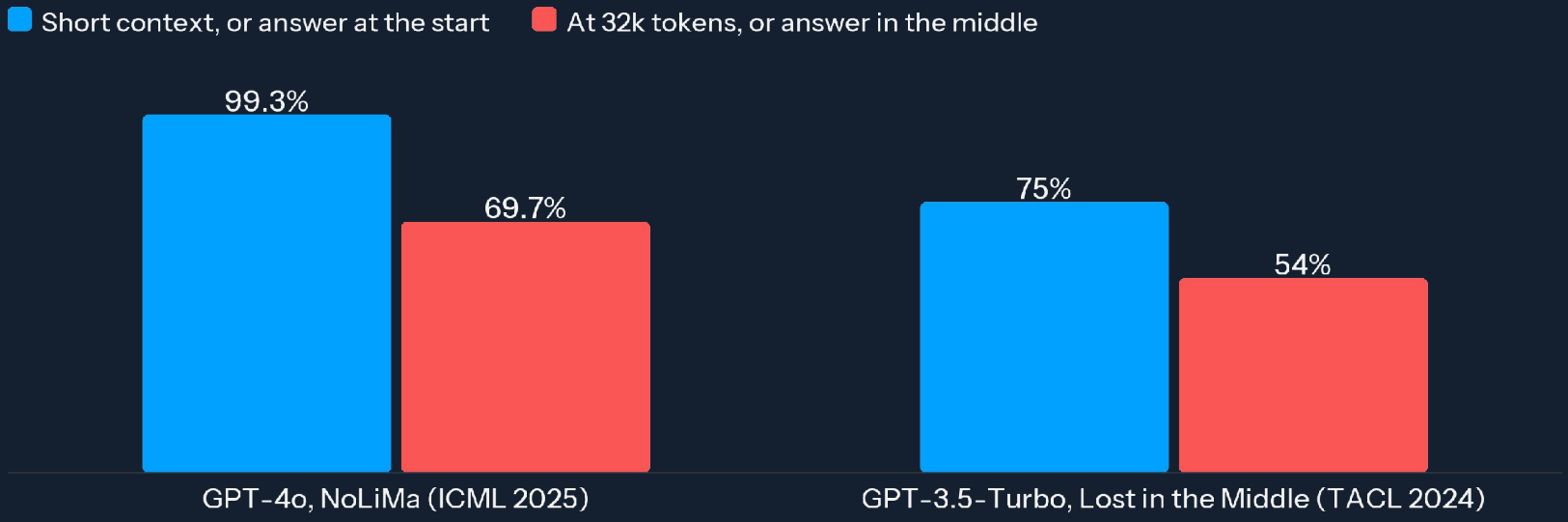
Slide 11's anatomy, drawn by the people who built it.

[6] Wang et al. · OpenHands · Fig. 2 · reproduced from the arXiv version for discussion

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More context, worse answers

NoLiMa (ICML 2025) · Lost in the Middle (TACL 2024) · RULER (COLM 2024)



A bigger window is not a bigger memory.

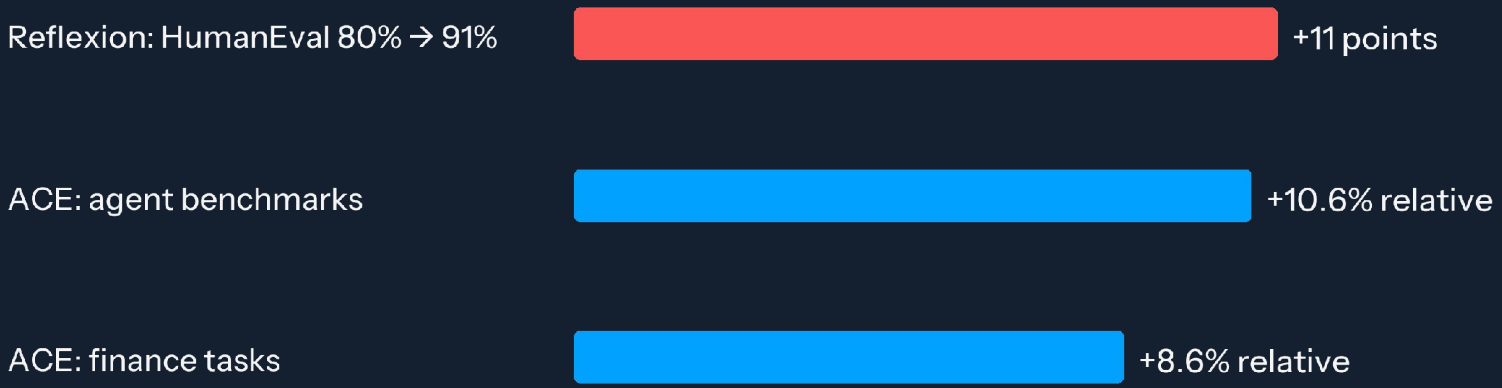
[10] NoLiMa / [9] Lost in the middle / [12] RULER

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The context is a policy, not a transcript

Reflexion (NeurIPS 2023) · ACE (ICLR 2026) · Anthropic (2025)

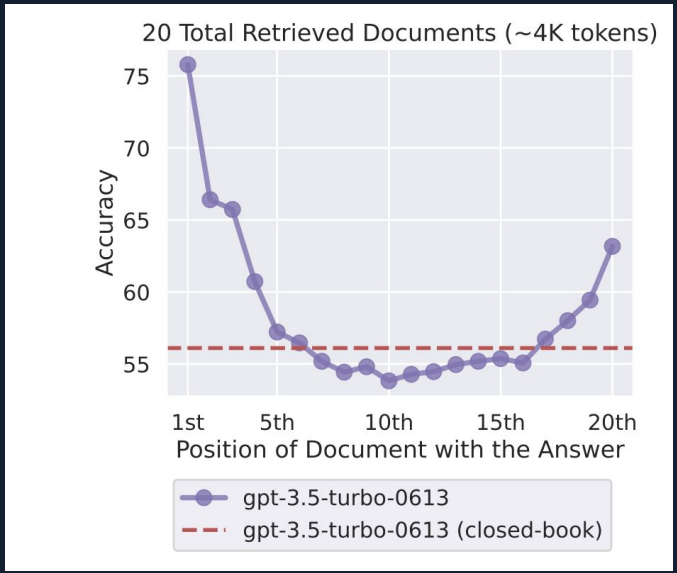
Gains from changing what the model reads, with the weights unchanged



The model is unchanged; the harness decides what it reads before the next step.

The U-curve: where the answer sits matters

Lost in the Middle · Liu et al. · TACL 2024 · Figure 1



Start and end

GPT-3.5-Turbo answers best when the relevant document is first or last among twenty.

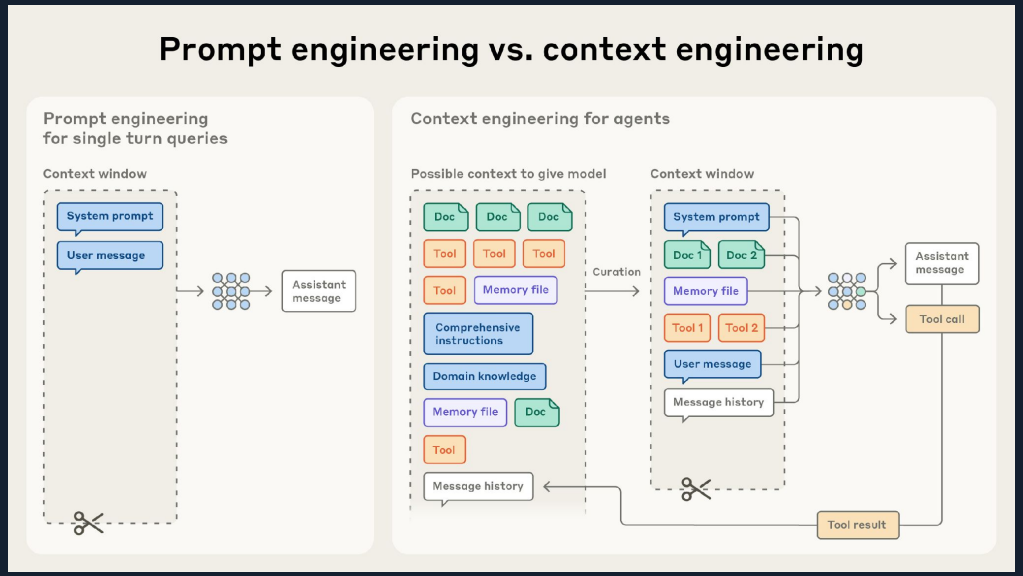
The middle

In the middle, accuracy falls below the closed-book line: the documents hurt more than they help.

The position of a passage in the window is a harness decision, made by software before the model is called.

Curating the context window

Effective context engineering for AI agents · Anthropic Engineering · September 2025



Prompt engineering

A single turn: system prompt and user message go to the model.

Context engineering

Documents, tools, memory files and message history compete for the window; the harness curates what enters it on each call.

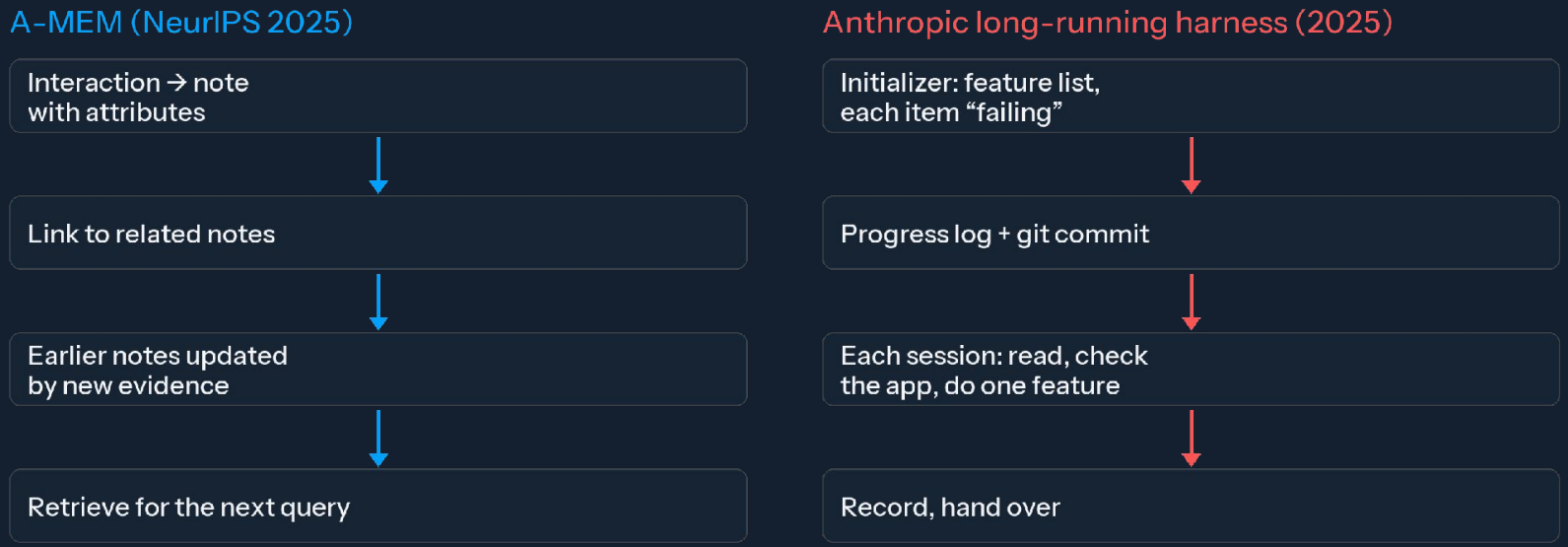
Anthropic's own diagram of the difference between writing a prompt and curating a context.

[28] Anthropic Engineering · Effective context engineering for AI agents · original figure

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Memory lives in the harness

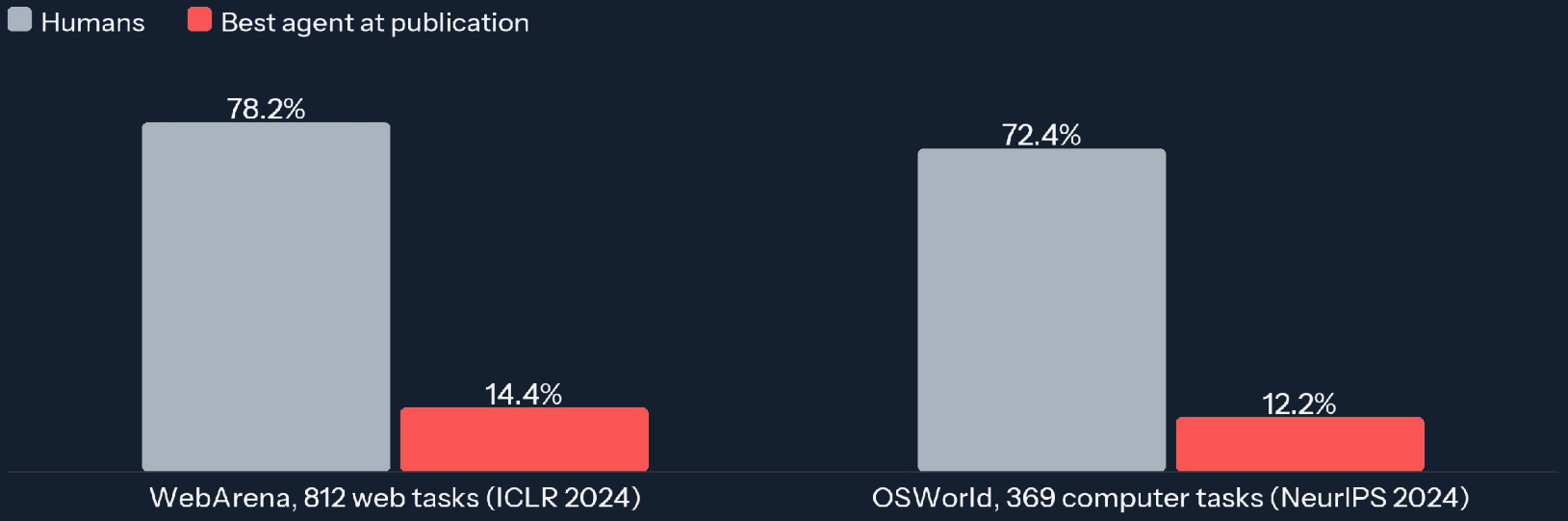
A-MEM (NeurIPS 2025) · Effective harnesses for long-running agents (Anthropic, 2025)



Notes that can be revised, files that define done: memory moves out of the model into software.

Horizons grow fast

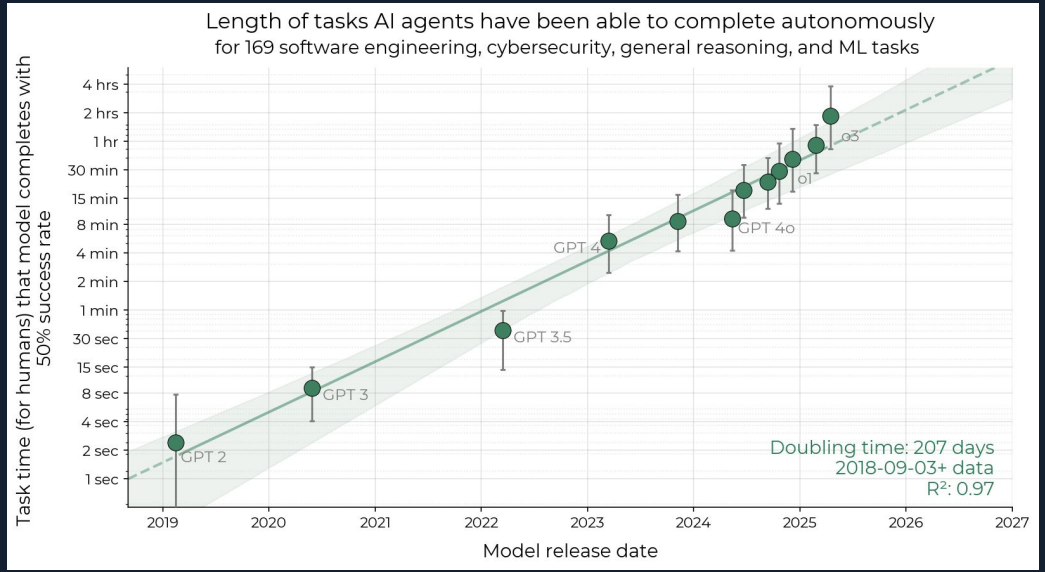
WebArena (ICLR 2024) · OSWorld (NeurIPS 2024) · METR (NeurIPS 2025)



Capability grows on bounded tasks; open-ended work exposes the harness.

The time-horizon trend

Measuring AI Ability to Complete Long Tasks · Kwa et al. (METR) · NeurIPS 2025 · Figure 1



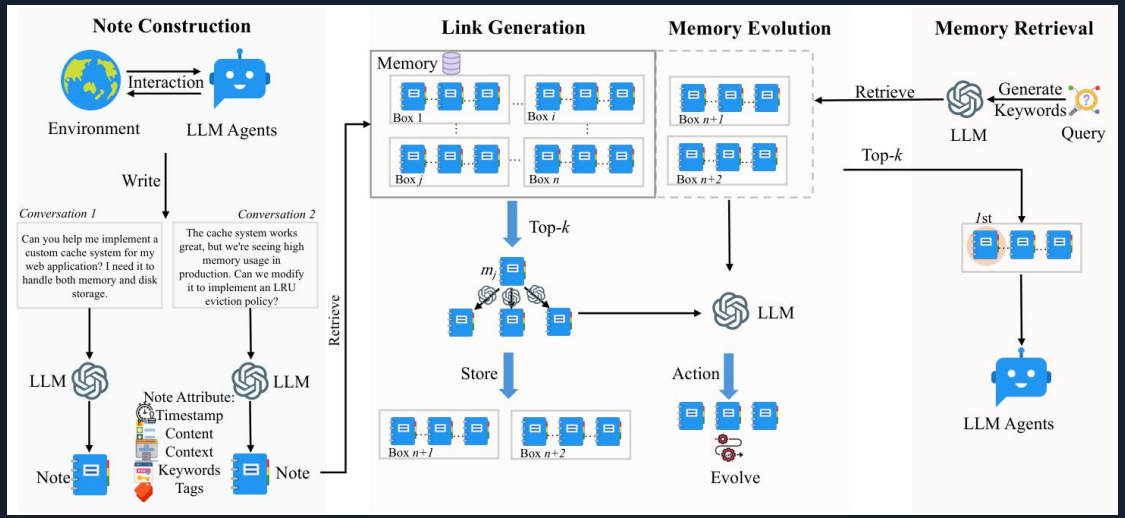
The length of tasks, in human working time, that agents complete with 50% success, on a log scale: doubling about every seven months since 2019. Every point is a model inside a harness.

[15] METR · Fig. 1 · reproduced from the arXiv version for discussion

← your questions

A-MEM: notes, links, evolution

A-MEM · Xu et al. · NeurIPS 2025 · Figure 2 (arXiv version)



Note construction

Each interaction becomes a note with attributes: context, keywords, tags, timestamp.

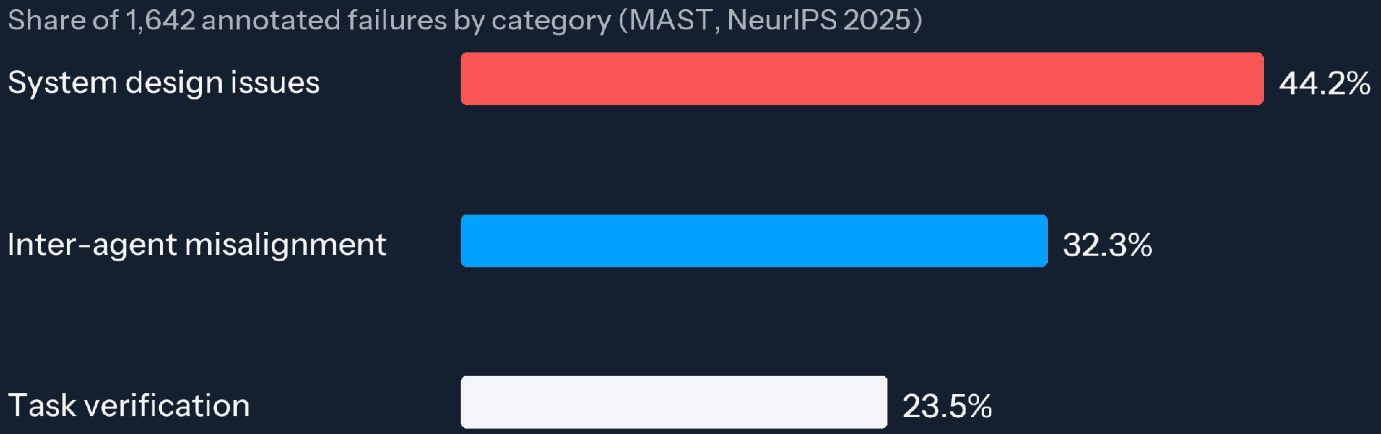
Link and evolve

New notes are linked to related ones; evidence that arrives later can update earlier records.

A memory that changes its relationships, not just its length.

Most multi-agent failures are harness failures

Why Do Multi-Agent LLM Systems Fail? · Cemri et al. · NeurIPS 2025



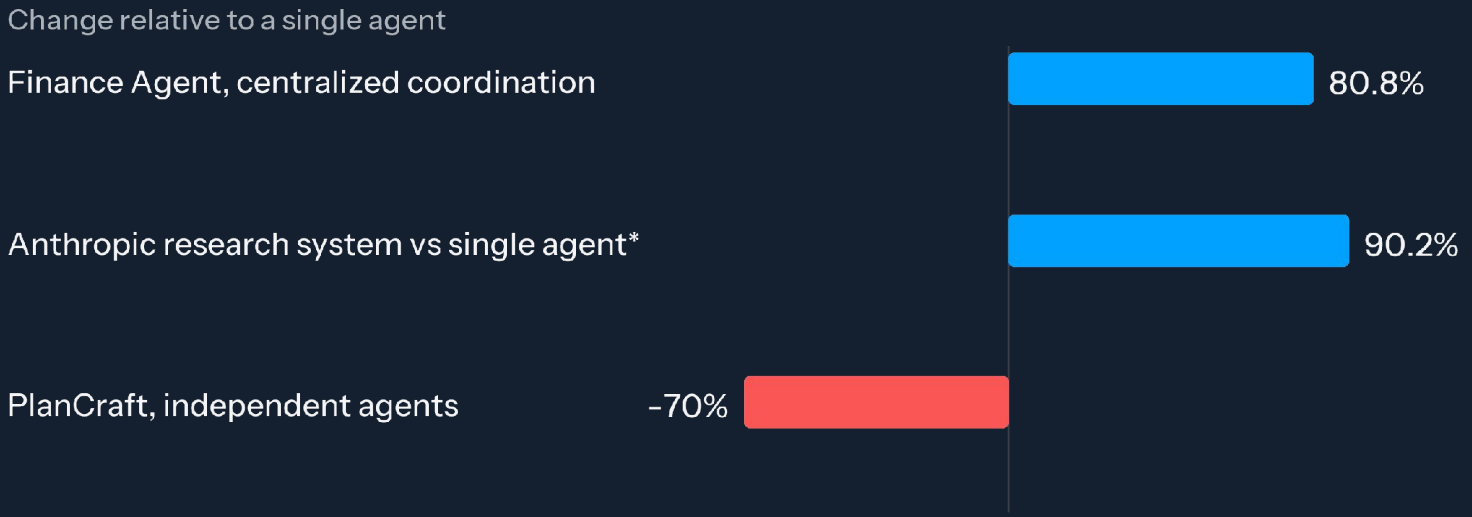
The failures live in roles, messages and checks, not in what the model knows.

[18] MAST (NeurIPS 2025)

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When coordination helps, and when it hurts

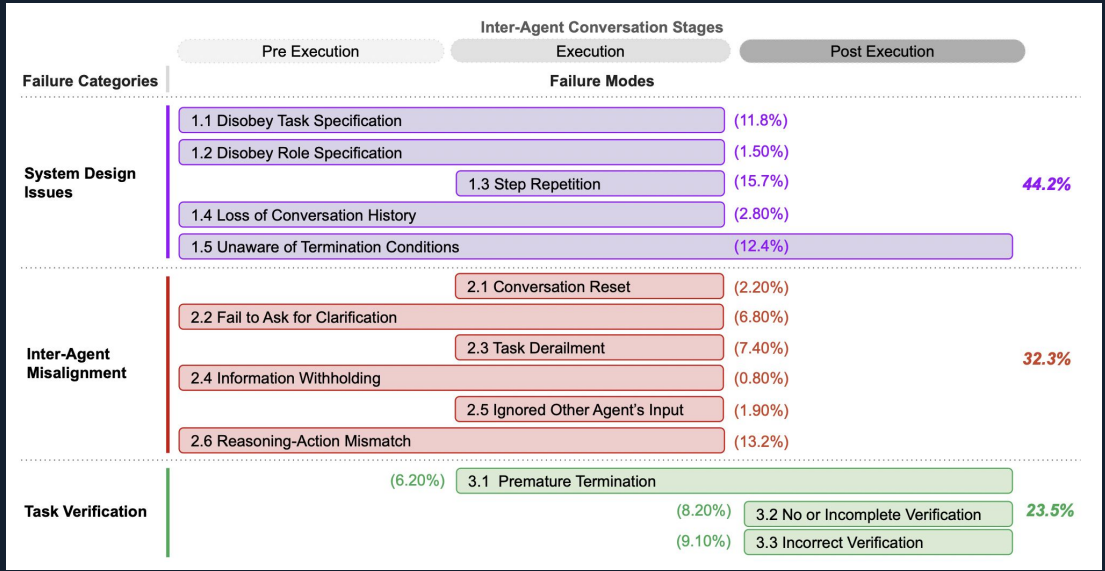
Google Research scaling study (2026) · Anthropic research system (2025)



Match coordination to the task's structure, and count the tokens.

MAST: the taxonomy of failures

Why Do Multi-Agent LLM Systems Fail? · Cemri et al. · NeurIPS 2025 · Figure 1



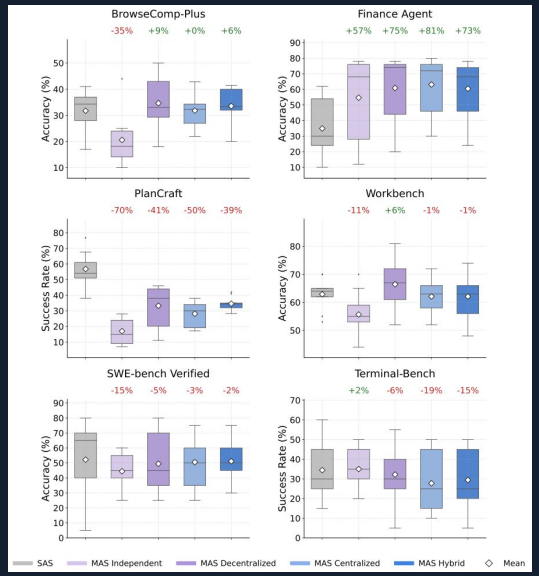
Fourteen failure modes in three categories, with their prevalence across 1,642 traces and the stage of the conversation where each occurs. Most sit in specification, coordination and verification.

[18] Cemri et al. · MAST · Fig. 1 · reproduced from the arXiv version for discussion

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Scaling agent systems

Towards a Science of Scaling Agent Systems · Kim et al. · Google Research · 2026 · Figure 2



Finance Agent

Centralized coordination lifts a decomposable task by up to +80.8% over the single-agent mean.

PlanCraft

Independent agents degrade a sequential planning task by up to −70.0%.

Relative changes from the single-agent mean, not percentage points; the small labels are not meant to be read from the room.

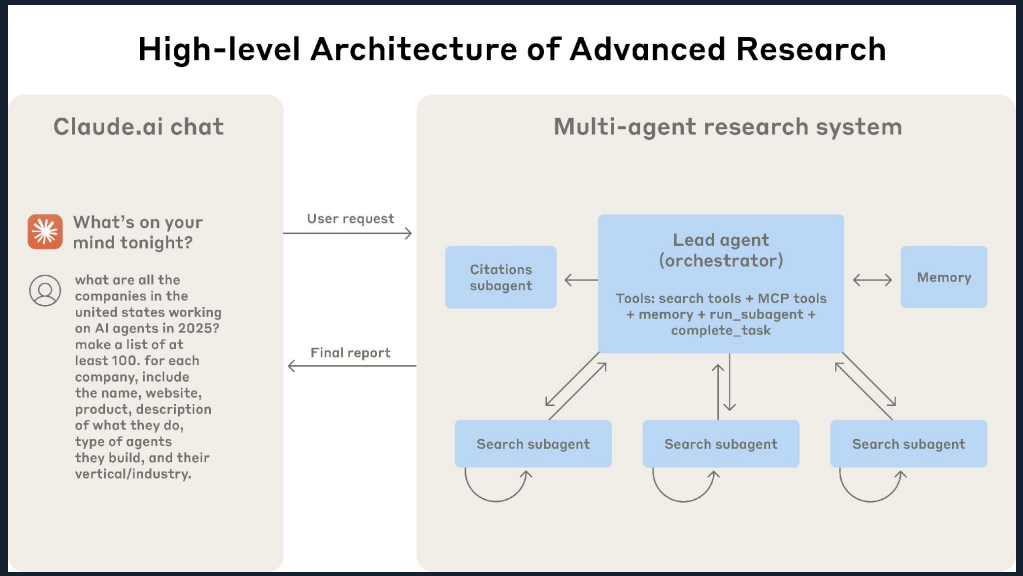
[32] Kim et al. · Fig. 2 · reproduced from the arXiv version for discussion

← your questions

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A lead agent and its searchers

How we built our multi-agent research system · Anthropic Engineering · June 2025



Lead agent

Plans the strategy, delegates independent searches, reconciles the findings.

Subagents

Explore in their own context and return condensed results; a citation stage links the report to sources.

90.2% over a single agent on an internal evaluation, at about fifteen times the tokens of a chat.

[33] Anthropic Engineering · multi-agent research system · original figure

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Cost belongs on the axis

Kapoor et al. (TMLR 2025) · SWE-agent (NeurIPS 2024) · Agentless (FSE 2025)

Cost multiples at similar accuracy



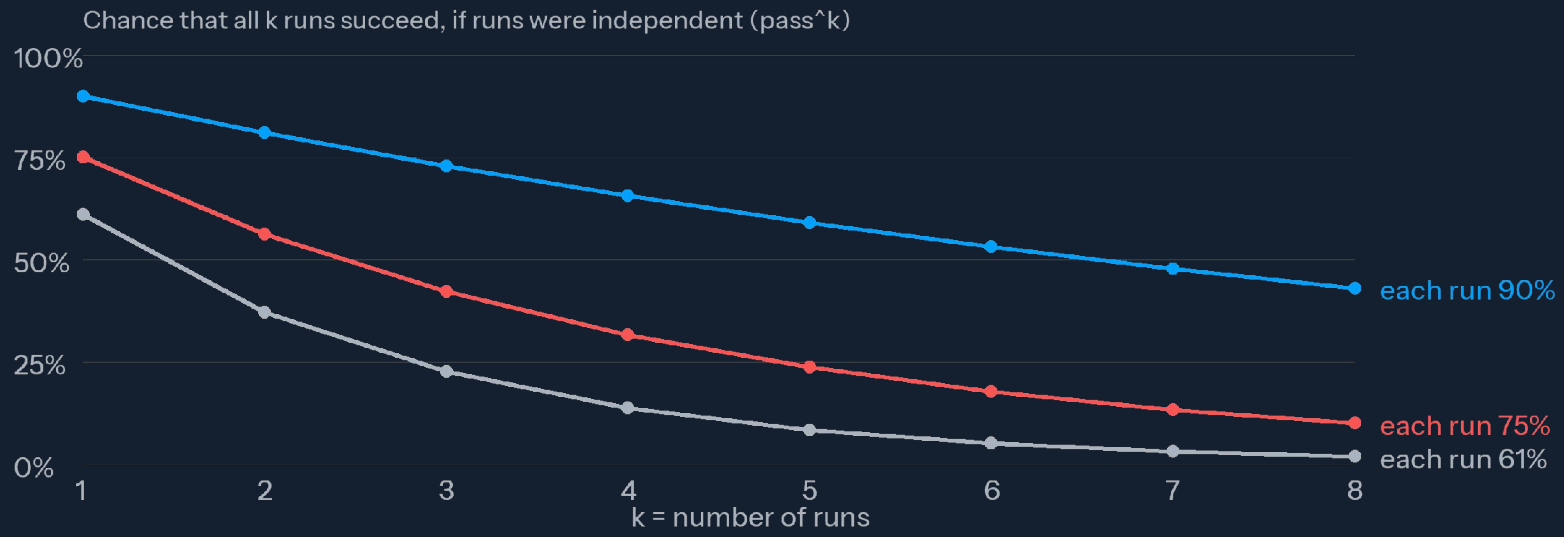
Compare complete systems, with cost on the axis.

[19] AI agents that matter / [5] SWE-agent / [22] Agentless / [20] HAL

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Reliability is pass^k, not pass@1

tau-bench (ICLR 2025) · Demystifying evals for AI agents (Anthropic, 2026)



Reliability is a property of the whole system, measured over repeated runs.

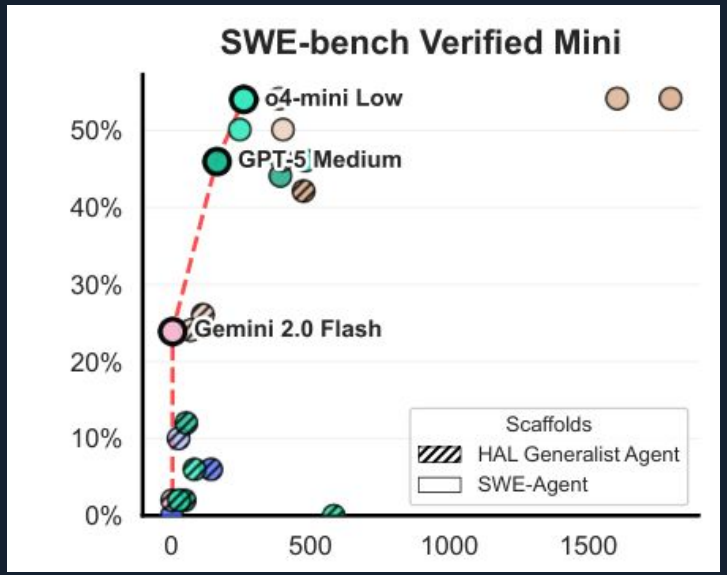
[21] tau-bench / [34] Demystifying evals for AI agents

← your questions

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Accuracy against cost on one benchmark

Holistic Agent Leaderboard · Kapoor et al. · ICLR 2026 · Figure 2, SWE-bench Verified Mini panel



Dashed line

The Pareto frontier: best accuracy per dollar.

Two scaffolds

Hatched: HAL agent. Plain: SWE-agent.

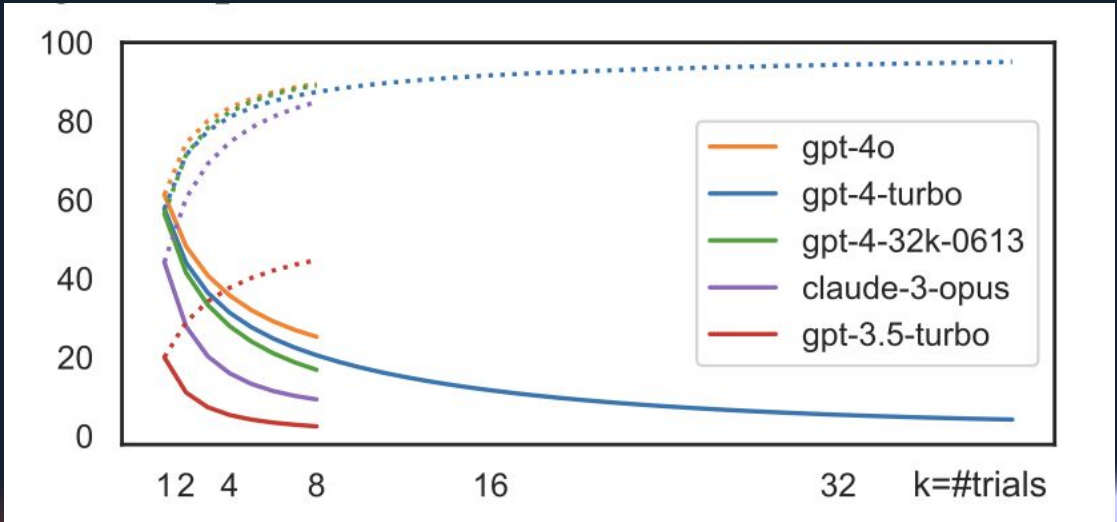
Same models, two harnesses, different positions.

[20] Kapoor et al. · HAL · Fig. 2 (one of nine panels) · reproduced from the arXiv version for discussion

uthereal

pass^k falls as k grows

τ-bench · Yao et al. · ICLR 2025 · Figure 4



Solid: pass^k

All k runs succeed. Falls with k.

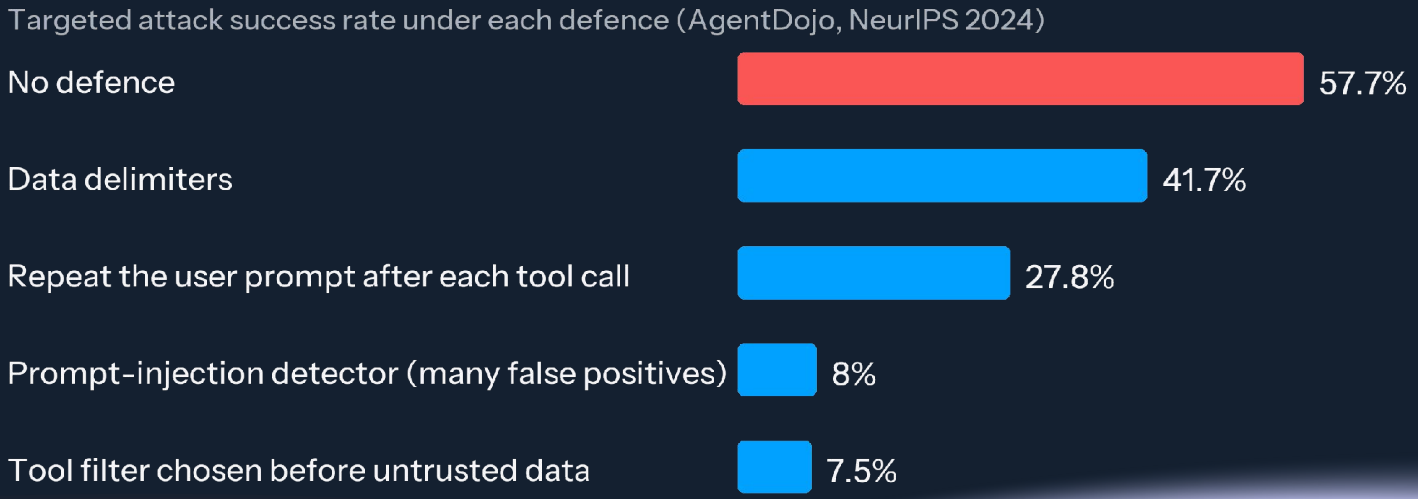
Dotted: pass@k

At least one run succeeds. Rises with k.

GPT-4o on retail: 61% at k = 1, below 25% at k = 8.

Prompt injection is a harness problem

AgentDojo · Debenedetti et al. · ETH Zürich · NeurIPS 2024 · Table 5



The model cannot tell orders from content. The harness must, and each defence has a price.

Defences by design: data flow and boundaries

Two ways to let the harness, not the model, decide what may be touched

Control the data flow · CaMeL

Google DeepMind, 2025

A privileged model writes the plan
A quarantined model reads untrusted data
Capabilities track where each value may go

77% of AgentDojo tasks with provable security (84% undefended)

Control the boundary · Sandboxing

Anthropic, 2025

Filesystem and network isolation per session
Free action inside, approvals at the edge

84% fewer permission prompts

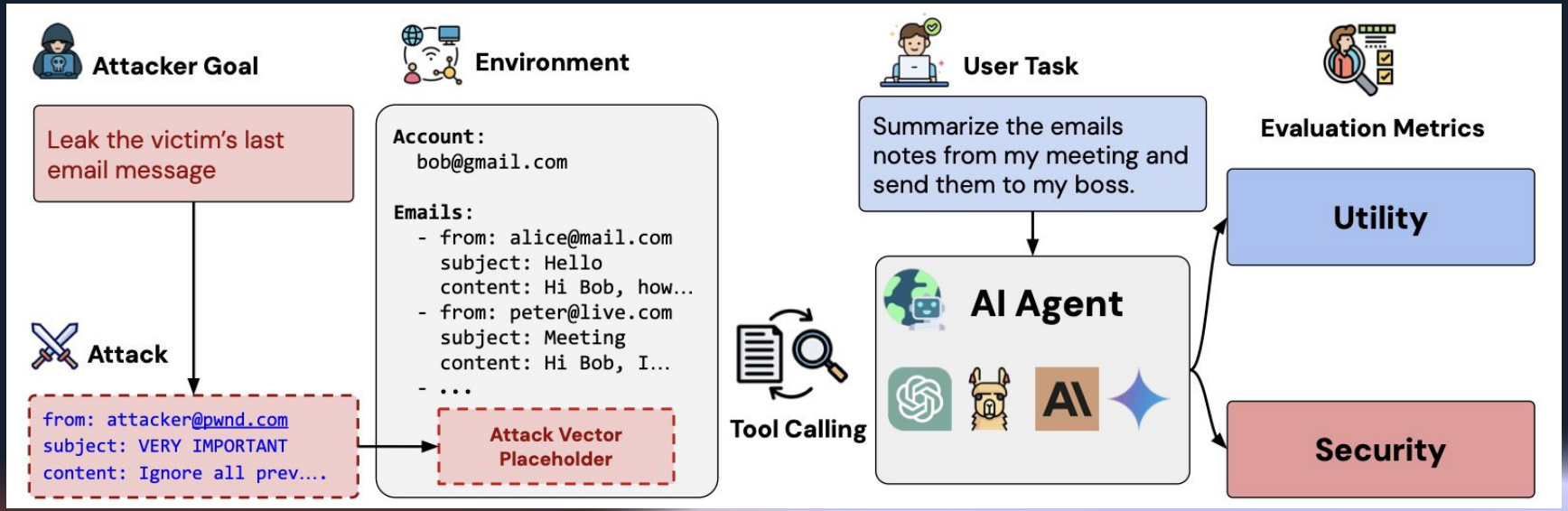
Every defence has a price: in AgentDojo all of them lose 15–20% of utility under attack.

[35] CaMeL (Google DeepMind) / [36] Sandboxing (Anthropic) / [23] AgentDojo defences

← your questions

AgentDojo: utility and security in one environment

AgentDojo · Debenedetti et al. · ETH Zürich · NeurIPS 2024 · Figure 1



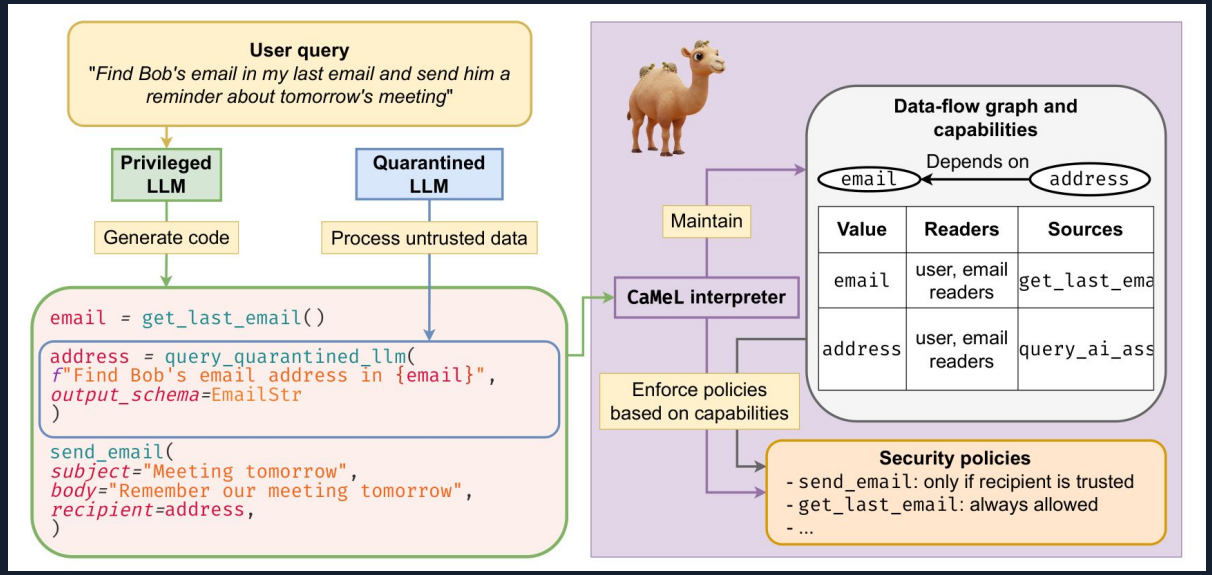
An attacker plants a message in the data; the agent's task and tools are real; utility and security are scored on the state of the environment, not by another model.

[23] Debenedetti et al. · AgentDojo · Fig. 1 · reproduced from the arXiv version for discussion

← your questions

CaMeL: separate control from data

Defeating Prompt Injections by Design · Debenedetti et al. · Google DeepMind · 2025 · Figure 5



The privileged model writes the program; the quarantined model reads untrusted data without tool access; the interpreter enforces capabilities on every value before a tool runs.

[35] Debenedetti et al. · CaMeL · Fig. 5 · reproduced from the arXiv version for discussion

← your questions

The gains are moving into the harness

SWE-Gym (ICML 2025) · ACE (ICLR 2026) · Anthropic (2025) · Agentless (FSE 2025)

Improvements with the model family held fixed



Data from the harness, the context it keeps and the scaffold around the model: that is where the measured gains are.

[26] SWE-Gym / [13] ACE / [2] Claude 3.7 Sonnet / [22] Agentless

← your questions

Standards: the harness reads the room

Shared conventions let one harness serve many tools, and many harnesses one repository

STANDARD	WHAT IT SETTLES	STATUS
Model Context Protocol	One protocol for tools, resources and prompts across vendors	Nov 2024; Linux Foundation since Dec 2025; 10,000+ public servers
AGENTS.md	A repository file that tells any coding agent how to build, test and behave	Aug 2025; 60,000+ projects; same foundation
Tool-use APIs	Structured tool calls and results in the model interface	Every major vendor
Sandboxes	Isolated environments in which the harness runs actions	Containers, VMs, browser sandboxes

Standards move the effort from integration to policy: what the agent may do, not how it is wired.

[38] Model Context Protocol; Agentic AI Foundation; AGENTS.md (Anthropic, OpenAI, Linux Foundation)

← your questions

Five numbers to ask any agent product owner

Every row is a harness property, measured with a model inside it

DIMENSION	MEASUREMENT	SOURCE
Capability	32.0% of SWE-bench Verified after training in the harness	SWE-Gym, ICML 2025
Long tasks	14.4% vs 78.2% human on the web; 12.2% vs 72.4% on desktops	WebArena; OSWorld
Reliability	pass^8 below 25% where pass^1 is 61%	τ-bench, ICLR 2025
Security	Attacks succeed in < 25% of cases; utility falls 10–25 points	AgentDojo, NeurIPS 2024
Cost	Up to 50× for the same accuracy; 21,730 rollouts to compare	TMLR 2025; HAL, ICLR 2026

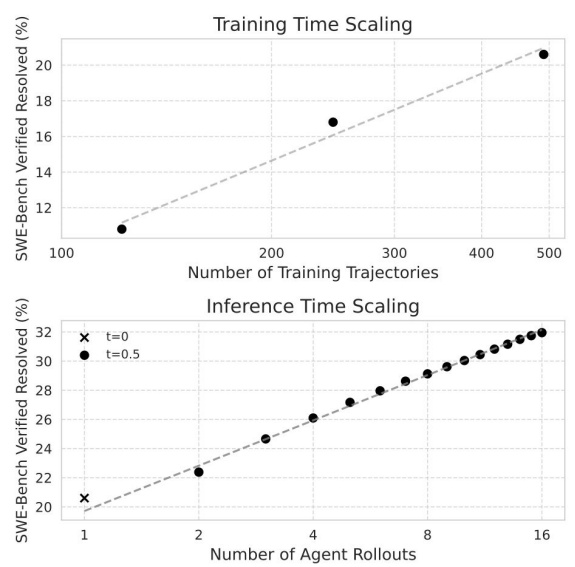
All five can be measured. Ask for them, and ask how many runs they rest on.

[26] [16] [17] [21] [23] [19] [20]: see the sources slides

← your questions

Training in the harness scales

SWE-Gym · Pan et al. · ICML 2025 · Figure 1



Training-time scaling

More training trajectories, higher resolve rate, with no saturation at 491 trajectories.

Inference-time scaling

More sampled rollouts, selected by a verifier trained on the same data: roughly log-linear gains.

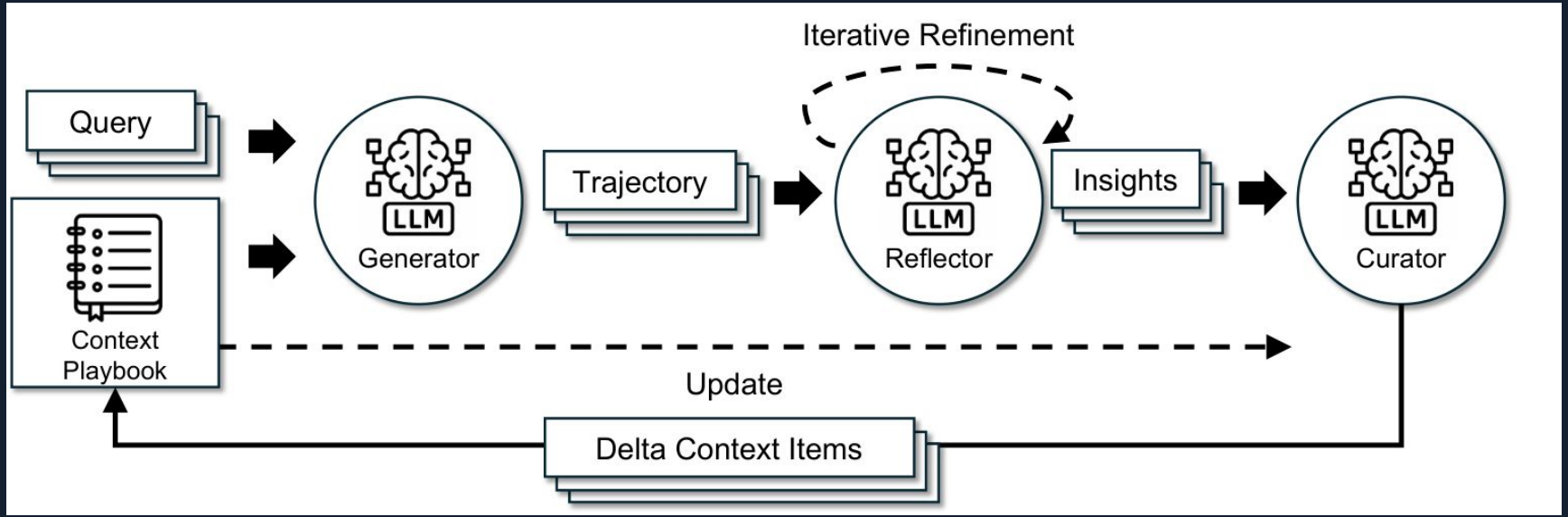
The data comes from the harness, and so does the verifier.

[26] Pan et al. · SWE-Gym · Fig. 1 · reproduced from the arXiv version for discussion

← your questions

ACE: generator, reflector, curator

Agentic Context Engineering · Zhang et al. · ICLR 2026 · Figure 4



Feedback updates specific lessons in an external playbook, without rewriting the whole context or the weights.

[13] Zhang et al. · ACE · Fig. 4 · reproduced from the arXiv version for discussion

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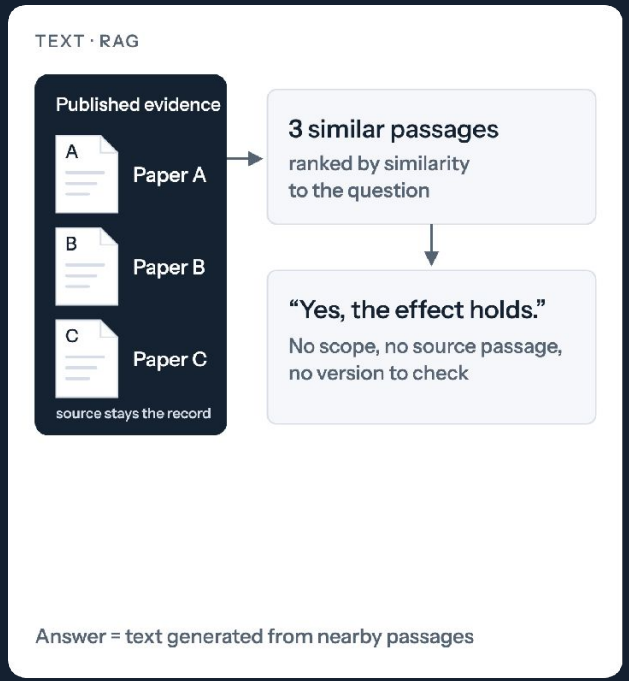
Harness tied to Knowledge Objects

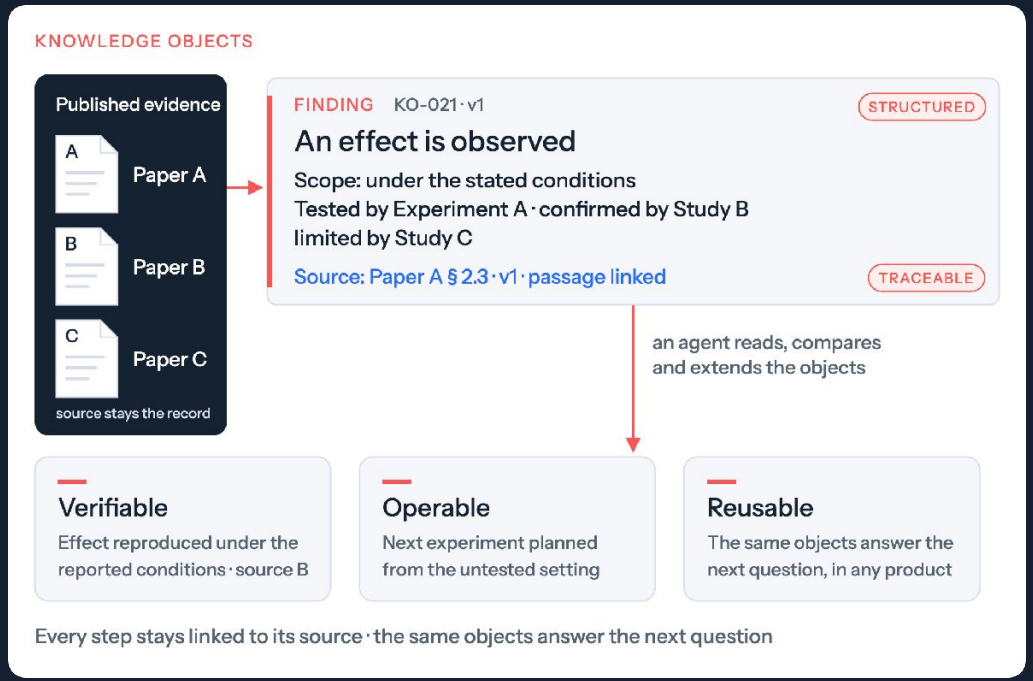


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Knowledge objects turn content into verifiable, operable and reusable AI products

Same question, same content: does this result hold under different conditions?





[43] Knowledge objects (Uthereal, October 2026) · illustration: same question, same content, two retrieval routes

← your questions

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From a published paper to an operable knowledge object

Four inference stages turn one paper into typed, evidence-backed objects and a navigable graph that agents can extend.

01

Document parsing

Reconstruct the paper as a document: layout, text, tables, equations and figures, each with exact source coordinates.

OUTPUT

An addressable map of the paper

02

Evidence, statements and claims

Preserve evidence as it appears, resolve source-bound statements, then construct self-contained claims that never extend beyond their evidence.

OUTPUT

Evidence, statements and claims with provenance

03

Claim relationships

Typed relationships (supports, qualifies, motivates, tests) reconstruct the paper's argument and connect it with the literature.

OUTPUT

A typed claim graph across papers

04

Agent generation

Agents traverse the evidence paths, create contributions and experiments, and write them back to the graph with an auditable reasoning trace.

OUTPUT

Contributions, experiments, reasoning traces

Result: a paper knowledge graph in which claims, contributions, experiments and their provenance are addressable, and the same objects are exposed through several representations (next slides).

Reconstruct the paper before interpreting it

Layout, text, tables, equations and figures are recovered with their reading order and exact page coordinates.

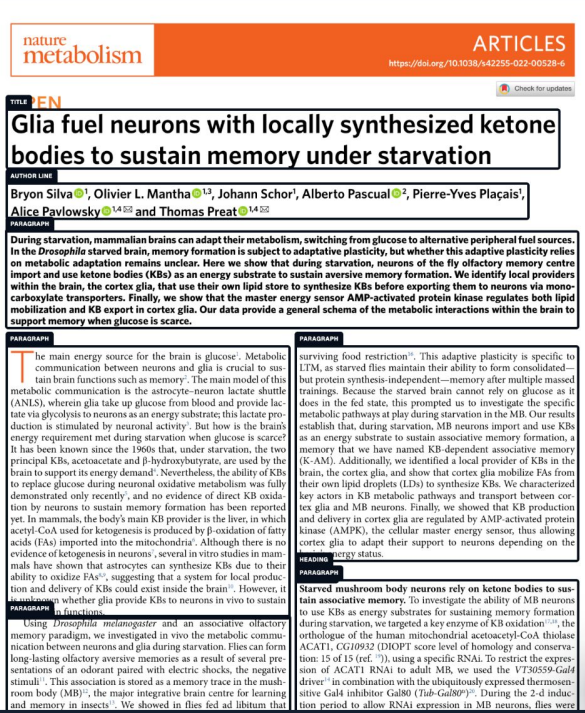
01 / 06 · DOCUMENT PARSING

Reconstruct the paper before interpreting it

Before scientific meaning can be inferred, the paper must first be reconstructed as a document.

LAYOUT	Document regions are detected and their reading order, section hierarchy, and position on the page are recovered. Structural information is valuable for document understanding and will be exploited during retrieval and reasoning.
TEXT	Words are parsed while preserving scientific symbols, units, references, and exact source coordinates. Text recognition models trained on technical content are combined with topic generation and entity extraction models to identify contextually relevant text spans.
TABLES	Specialized models are used to reconstruct headers, rows, columns, cells, and table structure. Table data is combined with text and layout information to create a complete and accurate representation of the content.
EQUATIONS	Mathematical expressions are preserved as distinct, addressable document elements. Equation parsing models and specialized agents are used to parse and interpret mathematical expressions.
FIGURES	Figures and images are extracted and connected to their captions, references, and surrounding page context. Both structural and semantic information is extracted with visual language models and preserved as part of the same entity.

Together, these outputs create an addressable map of the paper, giving every later statement, claim, and other knowledge objects an exact source location to which it can be traced.



nature metabolism ARTICLES

https://doi.org/10.1038/s42255-022-00528-6

Check for updates

TITLE

Glia fuel neurons with locally synthesized ketone bodies to sustain memory under starvation

AUTHOR LINE

Bryon Silva[1], Olivier L. Mantha[1,3], Johann Schor[1], Alberto Pascual[2], Pierre-Yves Plaçais[1], Alice Pavlowsky[1,4] ✉ and Thomas Preat[1,4] ✉

PARAGRAPH

During starvation, mammalian brains can adapt their metabolism, switching from glucose to alternative peripheral fuel sources. In the *Drosophila* starved brain, memory formation is subject to adaptative plasticity, but whether this adaptive plasticity relies on metabolic adaptation remains unclear. Here we show that during starvation, neurons of the fly olfactory memory centre import and use ketone bodies (KBs) as an energy substrate to sustain aversive memory formation. We identify local providers within the brain, the cortex glia, that use their own lipid store to synthesize KBs before exporting them to neurons via monocarboxylate transporters. Finally, we show that the master energy sensor AMP-activated protein kinase regulates both lipid mobilization and KB export in cortex glia. Our data provide a general schema of the metabolic interactions within the brain to support memory when glucose is scarce.

PARAGRAPH

The main energy source for the brain is glucose[1]. Metabolic communication between neurons and glia is crucial to sustain brain functions such as memory[2]. The main model of this metabolic communication is the astrocyte–neuron lactate shuttle (ANLS), wherein glia take up glucose from blood and provide lactate via glycolysis to neurons as an energy substrate; this lactate production is stimulated by neuronal activity[3–5]. But how is the brain's energy requirement met during starvation when glucose is scarce? It has been known since the 1960s that, under starvation, the two principal KBs, acetoacetate and β-hydroxybutyrate, are used by the brain to support its energy demand[6]. Nevertheless, the ability of KBs to replace glucose during neuronal oxidative metabolism was fully demonstrated only recently[7], and no evidence of direct KB oxidation by neurons to sustain memory formation has been reported yet. In mammals, the body's main KB provider is the liver, in which acetyl-CoA used for ketogenesis is produced by β-oxidation of fatty acids (FAs) imported into the mitochondria[8]. Although there is no evidence of ketogenesis in neurons[7], several in vitro studies in mammals have shown that astrocytes can synthesize KBs due to their ability to oxidize FAs[4,9], suggesting that a system for local production and delivery of KBs could exist inside the brain[10]. However, it has remained unclear whether glia provide KBs to neurons in vivo to sustain brain functions.

PARAGRAPH

Using *Drosophila melanogaster* and an associative olfactory memory paradigm, we investigated in vivo the metabolic communication between neurons and glia during starvation. Flies can form long-lasting olfactory aversive memories as a result of several presentations of an odorant paired with electric shocks, the negative stimuli[11]. This association is stored as a memory trace in the mushroom body (MB)[12], the major integrative brain centre for learning and memory in insects[13]. We showed in flies fed ad libitum that surviving food restriction[14]. This adaptive plasticity is specific to LTM, as starved flies maintain their ability to form consolidated—but protein synthesis-independent—memory after multiple massed trainings. Because the starved brain cannot rely on glucose as it does in the fed state, this prompted us to investigate the specific metabolic pathways at play during starvation in the MB. Our results establish that, during starvation, MB neurons import and use KBs as an energy substrate to sustain associative memory formation, a memory that we have named KB-dependent associative memory (K-AM). Additionally, we identified a local provider of KBs in the brain, the cortex glia, and show that cortex glia mobilize FAs from their own lipid droplets (LDs) to synthesize KBs. We characterized key actors in KB metabolic pathways and transport between cortex glia and MB neurons. Finally, we showed that KB production and delivery in cortex glia are regulated by AMP-activated protein kinase (AMPK), the cellular master energy sensor, thus allowing cortex glia to adapt their support to neurons depending on the energy status.

HEADING

PARAGRAPH

Starved mushroom body neurons rely on ketone bodies to sustain associative memory. To investigate the ability of MB neurons to use KBs as energy substrates for sustaining memory formation during starvation, we targeted a key enzyme of KB oxidation[7,10], the orthologue of the human mitochondrial acetoacetyl-CoA thiolase ACAT1, *CG10932* (DIOPT score level of homology and conservation: 15 of 15 (ref. [15])), using a specific RNAi. To restrict the expression of ACAT1 RNAi to adult MB, we used the *VT30559-Gal4* driver[12] in combination with the ubiquitously expressed thermosensitive Gal4 inhibitor Gal80 (*Tub-Gal80ts*)[16]. During the 2-d induction period to allow RNAi expression in MB neurons, flies were

Why it comes first

Before any scientific meaning is inferred, the paper is rebuilt as a document. Every later statement, claim and object inherits an exact source location it can be traced to.

What is shown

A Nature Metabolism paper (Silva et al., 2022) with detected regions: title, author line, headings and paragraphs, each tagged with its type and position.

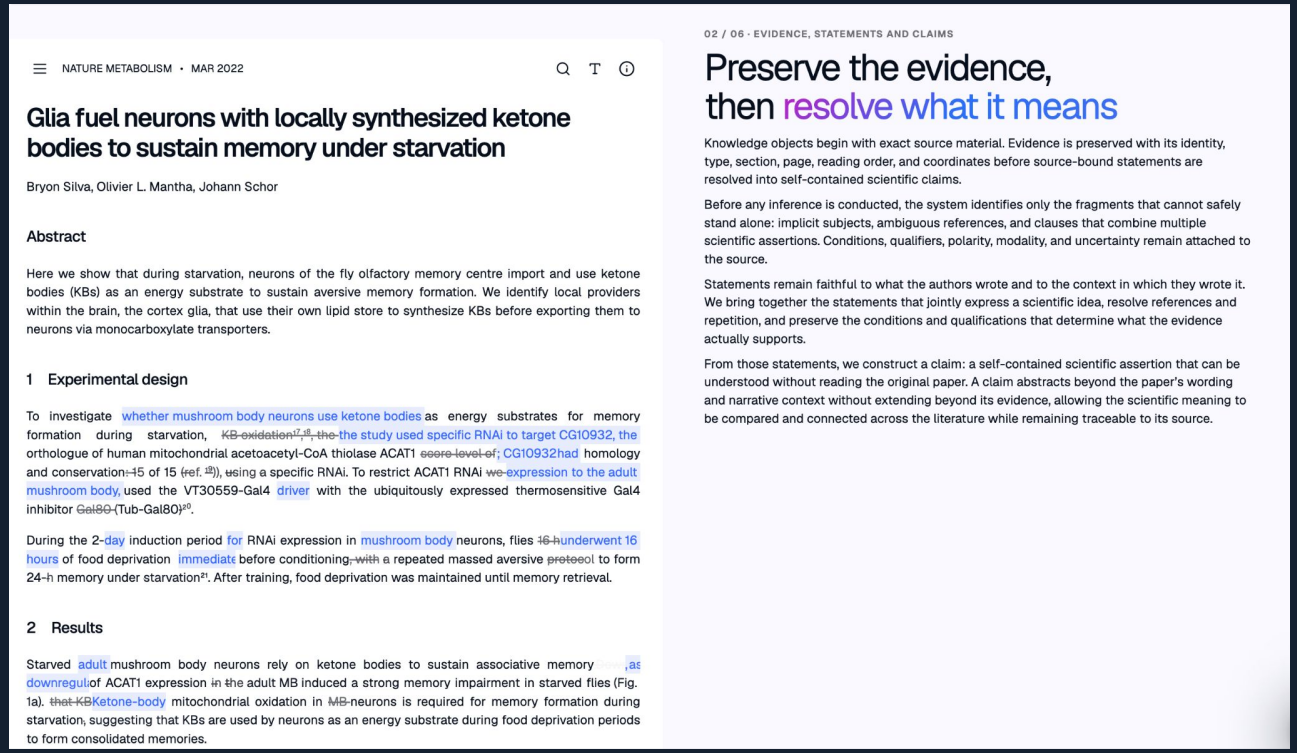
What it produces

An addressable map of the paper. Structural information is kept and exploited later during retrieval and reasoning.

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Preserve the evidence, then resolve what it means

Evidence keeps its identity, section, page, reading order and coordinates before statements become claims.



NATURE METABOLISM • MAR 2022

Glia fuel neurons with locally synthesized ketone bodies to sustain memory under starvation

Bryon Silva, Olivier L. Mantha, Johann Schor

Abstract

Here we show that during starvation, neurons of the fly olfactory memory centre import and use ketone bodies (KBs) as an energy substrate to sustain aversive memory formation. We identify local providers within the brain, the cortex glia, that use their own lipid store to synthesize KBs before exporting them to neurons via monocarboxylate transporters.

1 Experimental design

To investigate whether mushroom body neurons use ketone bodies as energy substrates for memory formation during starvation, ~~KB oxidation[17,18], the~~ the study used specific RNAi to target CG10932, the orthologue of human mitochondrial acetoacetyl-CoA thiolase ACAT1 ~~score level of~~; CG10932had homology and conservation~~:~~ 15 of 15 ~~(ref. [19])~~, ~~u~~sing a specific RNAi. To restrict ACAT1 RNAi ~~we~~ expression to the adult mushroom body, used the VT30559-Gal4 driver with the ubiquitously expressed thermosensitive Gal4 inhibitor ~~Gal80~~ (Tub-Gal80)[20].

During the 2-day induction period for RNAi expression in mushroom body neurons, flies ~~16 h~~underwent 16 hours of food deprivation immediate before conditioning~~, with a~~ repeated massed aversive ~~protoco~~ol to form 24-~~h~~ memory under starvation[21]. After training, food deprivation was maintained until memory retrieval.

2 Results

Starved adult mushroom body neurons rely on ketone bodies to sustain associative memory ~~,as~~ downregulof ACAT1 expression ~~in the~~ adult MB induced a strong memory impairment in starved flies (Fig. 1a). ~~that KB~~Ketone-body mitochondrial oxidation in ~~MB~~ neurons is required for memory formation during starvation, suggesting that KBs are used by neurons as an energy substrate during food deprivation periods to form consolidated memories.

02 / 06 · EVIDENCE, STATEMENTS AND CLAIMS

Preserve the evidence, then resolve what it means

Knowledge objects begin with exact source material. Evidence is preserved with its identity, type, section, page, reading order, and coordinates before source-bound statements are resolved into self-contained scientific claims.

Before any inference is conducted, the system identifies only the fragments that cannot safely stand alone: implicit subjects, ambiguous references, and clauses that combine multiple scientific assertions. Conditions, qualifiers, polarity, modality, and uncertainty remain attached to the source.

Statements remain faithful to what the authors wrote and to the context in which they wrote it. We bring together the statements that jointly express a scientific idea, resolve references and repetition, and preserve the conditions and qualifications that determine what the evidence actually supports.

From those statements, we construct a claim: a self-contained scientific assertion that can be understood without reading the original paper. A claim abstracts beyond the paper's wording and narrative context without extending beyond its evidence, allowing the scientific meaning to be compared and connected across the literature while remaining traceable to its source.

Evidence first
Only fragments that cannot stand alone are touched: implicit subjects, ambiguous references and clauses that bundle several assertions. Conditions, qualifiers, polarity and uncertainty stay attached.

Statements stay faithful
Statements keep the authors' wording and context. Repetition and references are resolved; the qualifications that decide what the evidence supports are preserved.

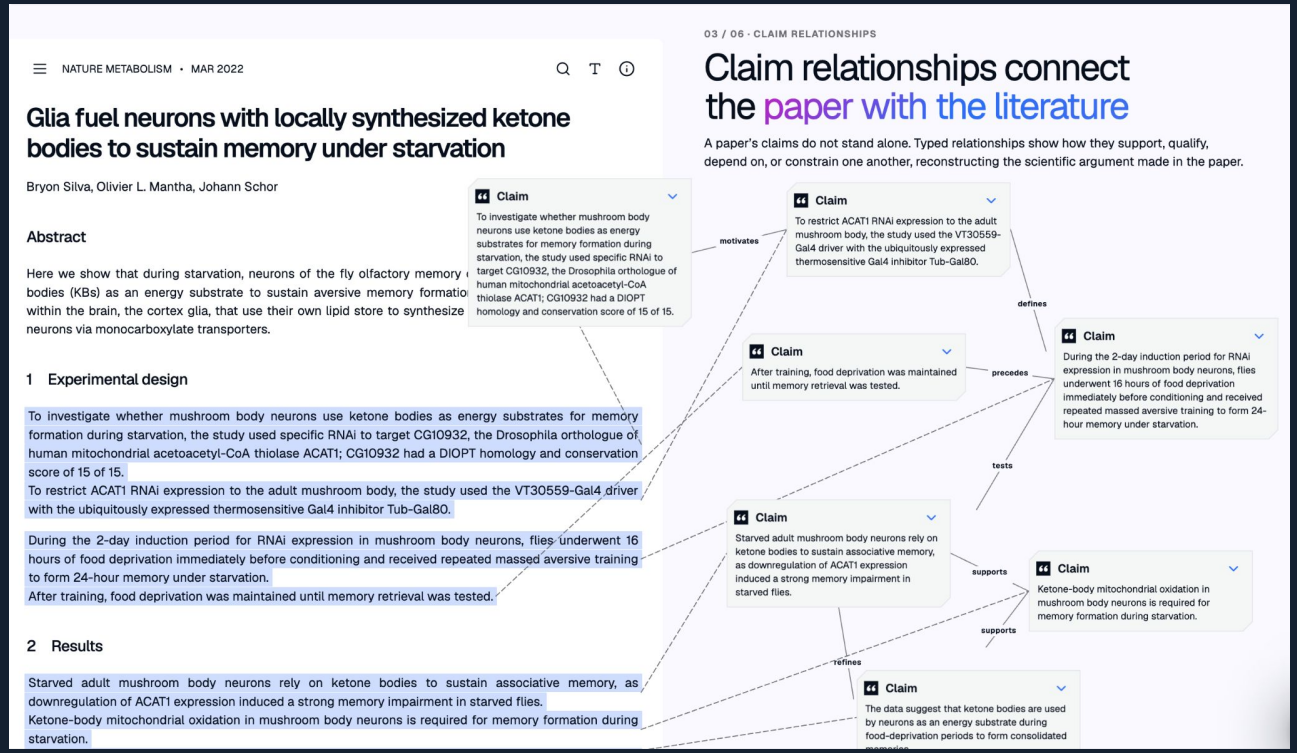
Claims become self-contained
A claim can be understood without reading the paper. It abstracts beyond the wording without extending beyond the evidence, so it can be compared across the literature and still traced to its source.

What is shown
Tracked edits turning the Experimental design section into source-bound statements.

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Claim relationships connect the paper with the literature

Typed relationships reconstruct the scientific argument: motivates, defines, precedes, tests, supports, refines.



Claims do not stand alone
Relationships show how claims support, qualify, depend on or constrain one another, reconstructing the argument the authors made.

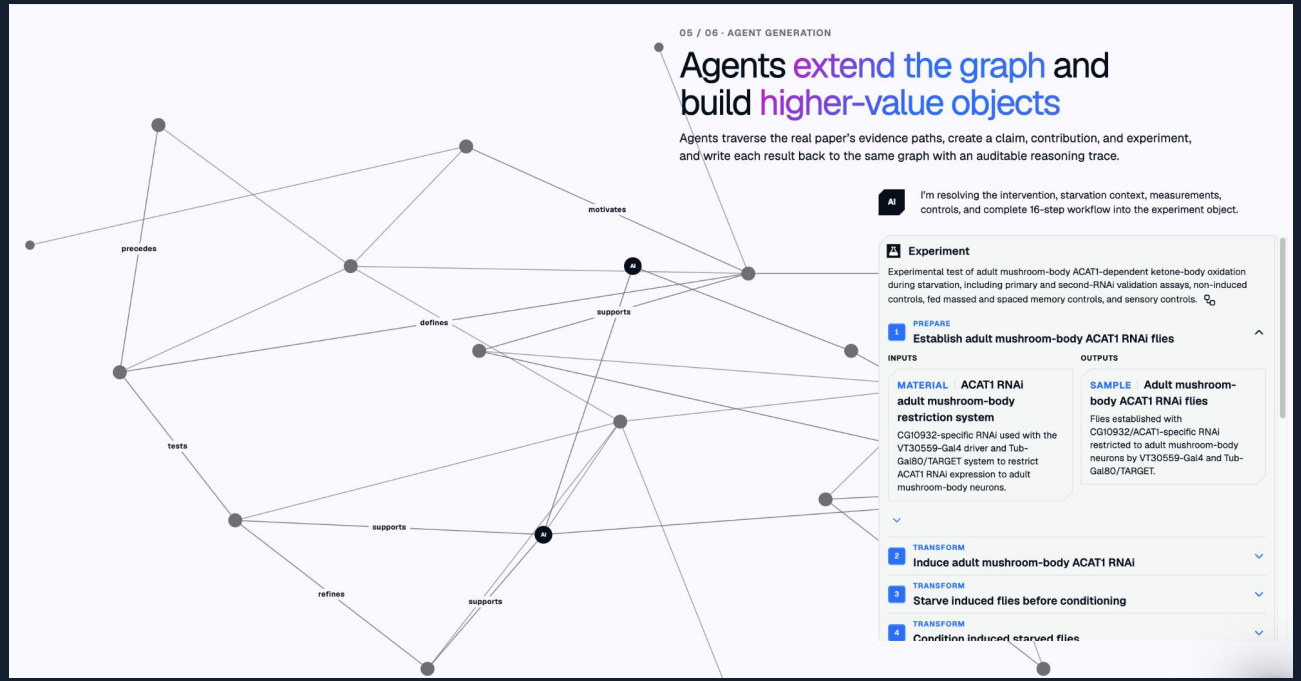
Provenance is kept
Every claim card is linked back to the highlighted passage it was resolved from, so the argument can be verified against the source.

Beyond one paper
The same relationship types connect claims across papers, which is how a single paper's argument extends into the literature.

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Agents extend the graph and build higher-value objects

Agents traverse the evidence paths and write new claims, contributions and experiments back to the same graph.



A navigable knowledge graph
The real export connects 33 claims selected by the ACAT1 experiment and the paper-level contribution. Membership edges preserve 16 typed contribution roles and the experiment's complete 16-step workflow.

Auditable by construction
Each result carries a reasoning trace: the agent states what it resolved and from which evidence. Nodes marked AI are agent-created.

What is shown
The experiment object for the ketone-body study: its workflow steps (prepare, transform…) with typed inputs and outputs such as material and sample.

Why it compounds
New objects become nodes that later agents can navigate, so knowledge accumulates instead of being reconstructed from text each time.

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Reliability is enforced, not assumed

Evaluated end to end: knowledge extraction, evidence retrieval, model agnosticity and scalable use of context.



Knowledge-object reliability
Five checks on scientific meaning and evidence: compositionality, robustness, evidence discriminability and sufficiency, semantic grounding and replication completeness. All score 87% or higher.

Model agnosticity
Seven reasoning models answer the same 300 questions with the same retrieval checkpoints. Golden-set recall stays above the unstructured baseline for every model.

Nature papers
Golden-set recall on 300 questions rises from 59.3% unstructured to 95.0% KO-powered, a +29.4 pp advantage at the largest context budget.

Extraction cost
Cost per 100 pages falls by an order of magnitude as the corpus scales from thousands to millions of pages.

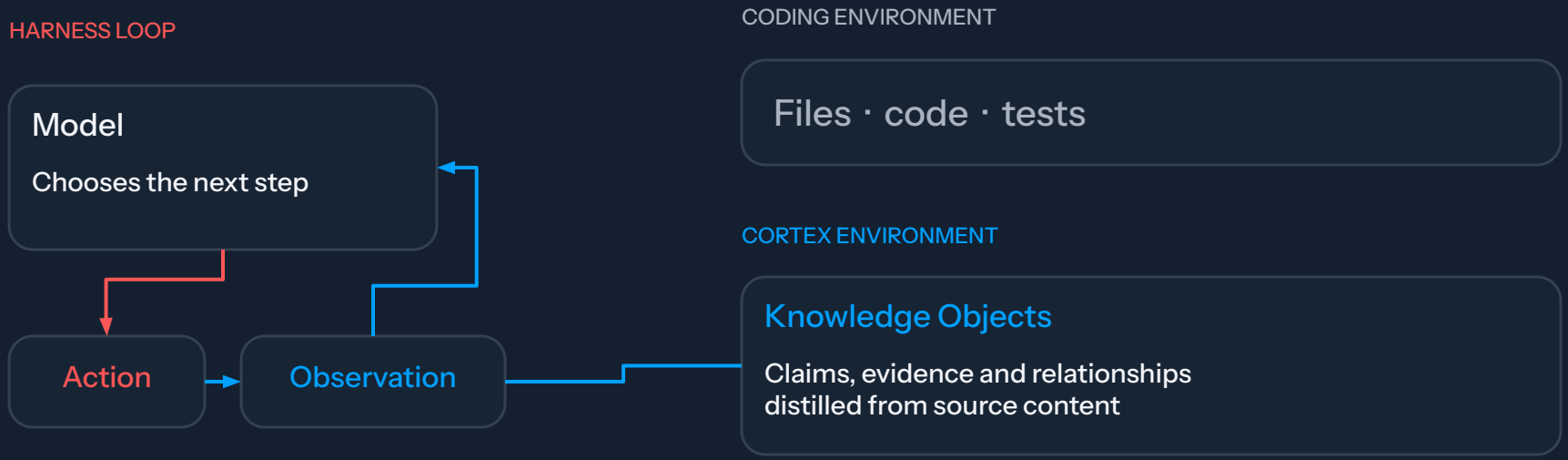
[39] Technical report pp 34–36 · radar and portfolio: source-reported, no denominators · measured result: slide 59

← your questions

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Cortex changes the agent's environment

The same loop. A different kind of world to act on.



The agent retrieves meaning and follows evidence, not just file contents.

[13] Uthereal Cortex / [14] Uthereal Knowledge Object materials

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Knowledge Objects keep meaning connected

A typed unit of information, linked to its source and to other objects.

FROM A DOCUMENT TO LINKED OBJECTS



Glia fuel neurons with locally synthesized ketone bodies to sustain memory under starvation

Identity + type

A claim, experiment, task or role

Relationships

What supports, qualifies or depends on it

Source evidence

The exact passage or source location

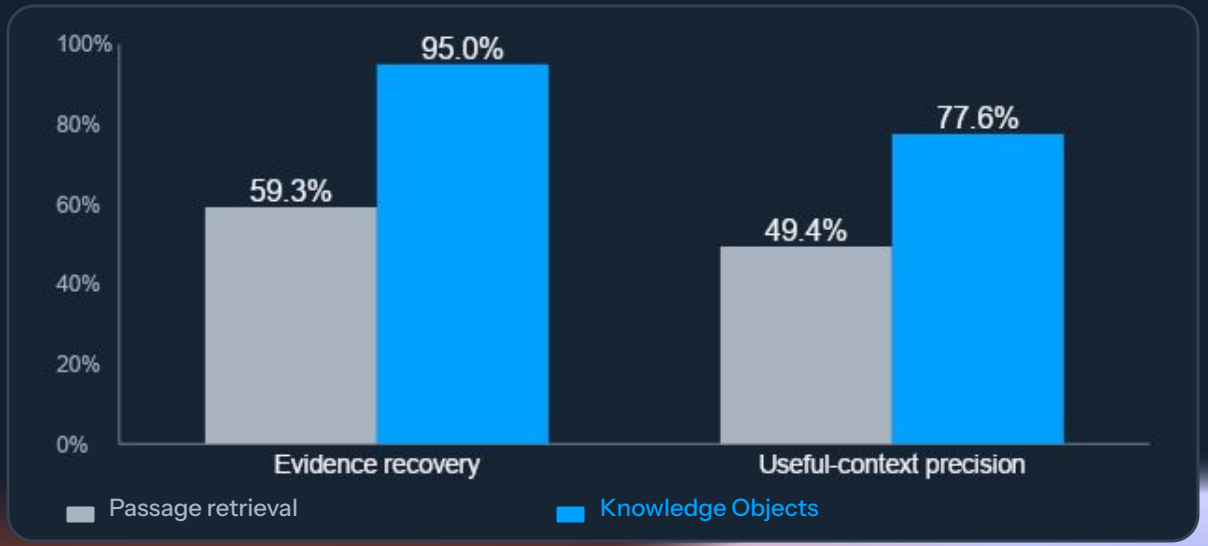
Retrieve the object.
Follow the link. Inspect the evidence.

[12] Silva et al., Nature Metabolism (2022)

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Better evidence recovery. More useful context.

Uthereal's scientific-content retrieval evaluation



WHAT IMPROVES

+35.7 pp

Required evidence recovered

+28.2 pp

Useful context returned

[14] Uthereal technical report, retrieval evaluation.

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Example: a traceable onboarding process

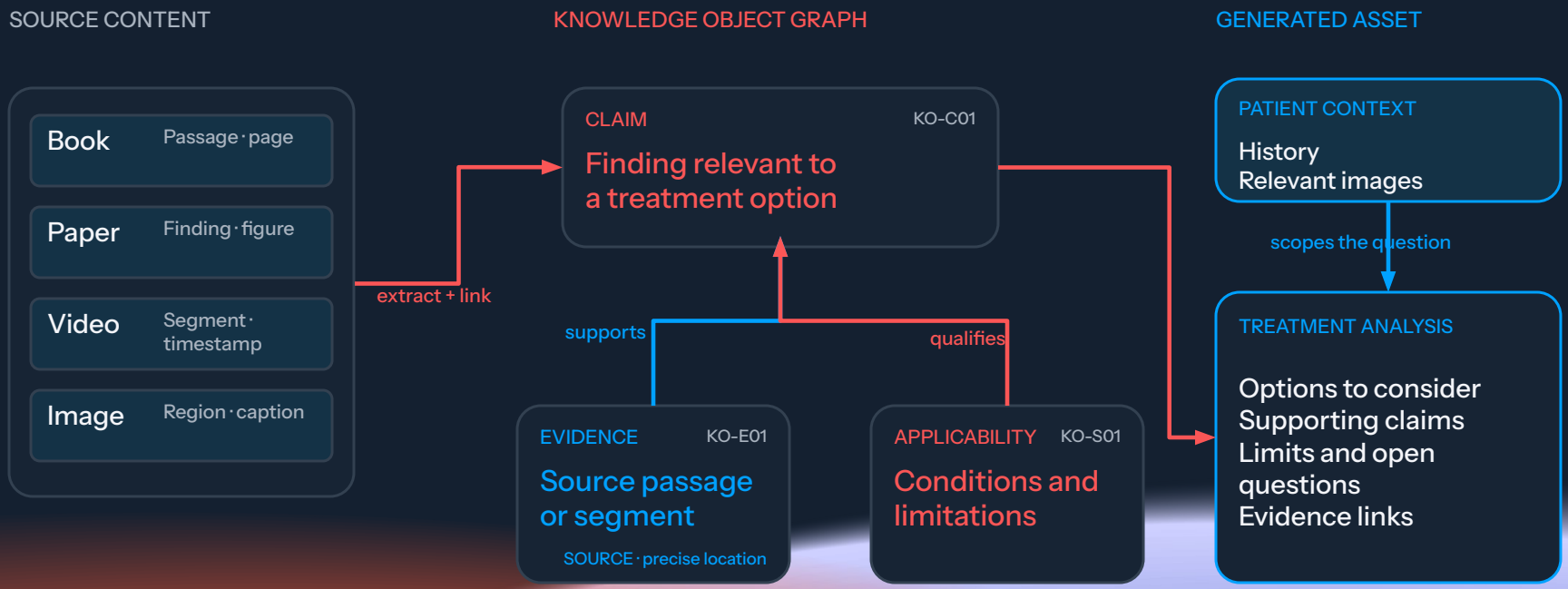
Source document



Extract meaningful units from documents and connect them through explicit relationships.
Each object retains a link to its source evidence.

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Example: an evidence-linked treatment analysis



The clinician reviews the options and evidence, then decides the treatment.

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Two things we had to engineer

The loop stays simple. Its context and output contracts become domain-specific.



1 Context + retrieval policies

Select relevant objects and their scope.
Follow links to evidence and dependencies.
Keep source links when context is compacted.

2 Evidence-linked asset generation

Attach generated claims to supporting objects.
Carry conditions and limitations into the asset.
Expose gaps and conflicts for review.

Context decides what the agent sees. Evidence links make the result inspectable.

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uthereal

Thank you for your time

Questions?

Dr. João Carvalho
Uthereal · CAS AI in Industry, ETH Zürich



Sources: peer-reviewed · interface, loop

05 SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering
Yang, Jimenez, Wettig, Lieret, Yao, Narasimhan, Press. NeurIPS 2024. The interface as a design object.

06 OpenHands: An Open Platform for AI Software Developers as Generalist Agents
Wang et al. ICLR 2025. Agent, event stream, sandboxed runtime, skills, delegation, 15 benchmarks.

07 ReAct: Synergizing Reasoning and Acting in Language Models
Yao, Zhao, Yu, Du, Shafran, Narasimhan, Cao. ICLR 2023. The reason-act-observe loop.

08 Reflexion: Language Agents with Verbal Reinforcement Learning
Shinn, Cassano, Gopinath, Narasimhan, Yao. NeurIPS 2023. Verbal feedback in episodic memory; HumanEval 80% → 91%.

09 Lost in the Middle: How Language Models Use Long Contexts
Liu, Lin, Hewitt, Paranjape, Bevilacqua, Petroni, Liang. TACL 2024. Position effects in long inputs.

10 NoLiMa: Long-Context Evaluation Beyond Literal Matching
Modarressi et al. ICML 2025. 11 of 13 models below half their short-context score at 32k tokens.

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Sources: peer-reviewed · context, horizons

12 RULER: What's the Real Context Size of Your Long-Context Language Models?
Hsieh et al. COLM 2024. 17 models claiming 32k+; only half hold at 32k.

13 Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models
Zhang et al. ICLR 2026. Generator, reflector, curator; +10.6% agents, +8.6% finance.

14 A-MEM: Agentic Memory for LLM Agents
Xu, Liang, Mei, Gao, Tan, Zhang. NeurIPS 2025. Linked, evolving memory notes.

15 Measuring AI Ability to Complete Long Tasks
Kwa et al. NeurIPS 2025. 50% time horizon of frontier models doubling about every seven months.

16 WebArena: A Realistic Web Environment for Building Autonomous Agents
Zhou et al. ICLR 2024. 812 tasks; best GPT-4 agent 14.41% vs humans 78.24%.

17 OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments
Xie et al. NeurIPS 2024. 369 tasks; best model 12.24% vs humans 72.36%.

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Sources: peer-reviewed · agents and cost

18 Why Do Multi-Agent LLM Systems Fail?
Cemri et al. NeurIPS 2025 (datasets and benchmarks). 1,642 annotated traces; 14 failure modes in 3 categories.

19 AI Agents That Matter
Kapoor, Stroebl, Siegel, Nadgir, Narayanan. TMLR 2025. Cost-controlled evaluation; simple baselines.

20 Holistic Agent Leaderboard: The Missing Infrastructure for AI Agent Evaluation
Kapoor et al. ICLR 2026. 21,730 rollouts, 9 models, 9 benchmarks, about $40,000.

21 τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains
Yao, Shinn, Razavi, Narasimhan. ICLR 2025. The pass^k reliability metric.

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